

357 homework questions

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April 2026

Week 1

1. Let a_n be a sequence in a metric space. Show that $a_n \rightarrow a$ iff every subsequence has a subsequence that converges to a .

Proof. $\Rightarrow$ direction is obvious. For the other direction, suppose that there exists an $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ there exists $n_N > N$ with $|a_{n_N} - a| > \varepsilon$. This is the definition of $a_n \not\rightarrow a$. However, then the subsequence $(a_{n_N})_N$ does not have a subsequence converging to $a \rightarrow \leftarrow$. $\square$

2. Let $X = C_0([0, 1])$. Show that $f_n(x) = x^n$ converges to 0 in the d_p metric with $p < \infty$.

Proof. Let $\varepsilon > 0$ be given. For $n \in \mathbb{N}$ satisfying $\frac{1}{np+1} < \varepsilon^p$,

$$\begin{aligned}\|f_n\|_p &= \left(\int_0^1 x^{np} \right)^{1/p} \\ &= \left(\frac{1}{np+1} \right)^{1/p} \\ &\leq \varepsilon\end{aligned}$$

$\square$

3. X as in the previous problem. Show that the function $F : X \rightarrow \mathbb{R}$ given by $F(f) = f(0)$ is continuous with respect to the d_∞ metric but not with respect to d_p metric for $p \in [1, \infty)$.

Proof. Given $\varepsilon > 0$, we have $\|f - g\|_\infty < \varepsilon \implies |f(0) - g(0)| < \varepsilon$ automatically. So F is continuous wrt d_∞ metric. However, $f(x) = 0$ and $g_n(x) = 1 - x^n$. By the previous part, $g_n \rightarrow 0$ so for any $\delta > 0$, there exists g_n with $\|g_n\|_p < \delta$ but we always have $|g_n(0) - f(0)| = 1$ so F is not continuous at 0. $\square$

4. **Young's Inequality + Holder's Inequality + Minkowski Inequality** Prove that if $x, y > 0$, then $xy \leq \frac{1}{p}x^p + \frac{1}{q}y^q$ whenever $\frac{1}{p} + \frac{1}{q} = 1$.

Proof. For convex functions, $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$. Set $x = e^{\frac{1}{p}X}, y = e^{\frac{1}{q}Y}$. Then $xy \leq \frac{1}{p}e^X + \frac{1}{q}e^Y = \frac{1}{p}x^p + \frac{1}{q}y^q$. $\square$

Prove that $\|fg\|_1 \leq \|f\|_p \cdot \|g\|_q$ whenever $\frac{1}{p} + \frac{1}{q} = 1$ and show $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Proof. WLOG, $\|f\|_p = \|g\|_q = 1$, then using Young's

$$\begin{aligned}
\|fg\|_1 &= \int |fg| \\
&\leq \frac{1}{p} \int |f|^p + \frac{1}{q} \int |g|^q \\
&= \frac{1}{p} \|f\|_p^p + \frac{1}{q} \|g\|_q^q \\
&= 1 \\
&= \|f\|_p \cdot \|g\|_q
\end{aligned}$$

This proves Holder's. Finally, for Minkowski

$$\begin{aligned}
\|f + g\|_p^p &= \int |f + g|^p \\
&= \int |f + g| \cdot |f + g|^{p-1} \\
&\leq \int |f| \cdot |f + g|^{p-1} + \int |g| \cdot |f + g|^{p-1} \\
&\leq \|f\|_p \cdot \|f + g\|_p^{p-1} + \|g\|_p \cdot \|f + g\|_p^{p-1} \quad \text{by Holder's} \\
&= (\|f\|_p + \|g\|_p) \cdot \|f + g\|_p^{p-1}
\end{aligned}$$

Hence $\|f + g\|_p \leq \|f\|_p + \|g\|_p$. □

5. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous with bounded support prove that $\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty$.

Proof. Let $\text{supp } f = [a, b]$, $|f| \leq M$. Given any $\varepsilon > 0$, let $I = \{x \in [a, b] : |f(x)| > M - \varepsilon\}$. Then, by continuity of f , I is an open set and since M is the supremum I is non-empty. Let $p \in I$ then $(p - \frac{t}{2}, p + \frac{t}{2}) \in I$ for some $t \in \mathbb{R}$. Then we have

$$\begin{aligned}
\left(\int_a^b |f|^p \right)^{1/p} &= \left(\int_I |f|^p + \int_{[a,b] \setminus I} |f|^p \right)^{1/p} \\
&\geq t^{1/p} (M - \varepsilon) + 0 \\
&\xrightarrow{p \rightarrow \infty} M - \varepsilon
\end{aligned}$$

Since this is true for any ε , $\lim_{p \rightarrow \infty} \|f\|_p = M = \|f\|_\infty$. □

6. Let a_n be a sequence. Prove that $p \rightarrow \|a\|_p$ is decreasing in p .

Proof. It suffices to show $\log(\|a\|_p)$ is decreasing in p since $\log x$ is an increasing function. So $\|a\|_p \leq \|a\|_q \iff \log \|a\|_p \leq \log \|a\|_q$. Now we show $\frac{d}{dp} \log \|a\|_p \leq 0$.

$$\frac{d}{dp} \log \|a\|_p \leq 0 \iff \sum |a_n|^p \ln |a_n|^p \leq \left(\sum |a_n|^p \right) \log \left(\sum |a_n|^p \right)$$

which is true since for every term in the sum,

$$|a_n|^p \ln |a_n|^p \leq |a_n|^p \cdot \ln \left(\sum |a_n|^p \right)$$

□

7. For any $1 \leq p < q \leq \infty$, find a sequence of elements $a^{(n)} \in \ell^1$ that converges to 0 in ℓ^q but not in ℓ^p .

Proof. Consider the sequence

$$a^{(n)} = (n^{-1/p}, \dots, n^{-1/p}, 0, 0, \dots)$$

where $n^{-1/p}$ occurs n many times. This sequence converges to 0 in ℓ^q but not in ℓ^p .

$$\begin{aligned} \|a^{(n)}\|_q &= n^{1-\frac{q}{p}} \xrightarrow{n \rightarrow \infty} 0 \\ \|a^{(n)}\|_p &= 1 \end{aligned}$$

□

9. Show that $D(x, y) = \frac{d(x, y)}{1+d(x, y)}$ is a metric space and $d(x_n, x) \rightarrow 0 \iff D(x, x_n) \rightarrow 0$.

Proof. The triangle inequality for D follows from the fact that $\frac{x}{1+x}$ is an increasing function and d satisfies triangle inequality. If $d(x_n, x) \rightarrow 0$ then obviously $D(x_n, x) \rightarrow 0$. For the other direction, notice that $d(x_n, x) < (1 + d(x_n, x))\varepsilon \implies d(x_n, x) \leq \frac{\varepsilon}{1-\varepsilon}$. □

13. (Pugh) Let $f : M \rightarrow N$ be such that whenever a sequence (p_n) converges in M , the sequence $(f(p_n))$ converges in N . Show that f is continuous.

Proof. Suppose f is not continuous at some point $x \in M$. Then there exists $\varepsilon > 0$ such that for every $n \in \mathbb{N}$, there is a point $x_n \in (x - \frac{1}{n}, x + \frac{1}{n})$ with $|f(x) - f(x_n)| > \varepsilon$. Then the sequence $(x_1, x, x_2, x, x_3, x, \dots, x_n, x, x_{n+1}, \dots)$ converges to x but $(f(x_1), f(x), f(x_2), f(x), \dots)$ does not converge as it is not Cauchy in $\mathbb{R}$. □

37. (Pugh) Construct a function that is continuous only at points of $\mathbb{Z}$.

Proof.

$$f(x) = \begin{cases} x - \lfloor x \rfloor & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

□

Week 2

1. Show that ℓ^∞ is complete.

Proof. Let $a^{(n)} \in \ell^\infty$ form a Cauchy sequence. Then, for fixed $n \in \mathbb{N}$, the sequence $\{a_n^{(m)}\}_{m=1}^\infty$ is Cauchy since given $\varepsilon > 0$, for large enough k, ℓ

$$|a_n^{(k)} - a_n^{(\ell)}| \leq \|a^{(k)} - a^{(\ell)}\|_\infty < \varepsilon$$

As $\mathbb{R}$ is complete, we can set $b_n = \lim_{m \rightarrow \infty} a_n^{(m)}$ and form the sequence $b = (b_n)$. Then, $a^{(n)} \rightarrow b$ in ℓ^∞ . To see this, let $\varepsilon > 0$. Then there exists $n \in \mathbb{N}$ such that $|a_k^{(n)} - a_k^{(m)}| < \varepsilon/2$ for all $n, m > N$, for all k . Then we have that $|b_k - a_k^{(n)}| \leq \varepsilon/2$. Otherwise, we can let $r = |b_k - a_k^{(n)}|$ and pick $p > N$ such that $|b_k - a_k^{(p)}| < (r - \varepsilon/2)$ and $|a_k^{(p)} - a_k^{(n)}| < \varepsilon/2$. But then

$$r = |b_k - a_k^{(n)}| \leq |b_k - a_k^{(p)}| + |a_k^{(p)} - a_k^{(n)}| < (r - \varepsilon/2) + \varepsilon/2 = r \implies r < r$$

Hence $|b_k - a_k^{(n)}| \leq \varepsilon/2$ indeed. Since $a^{(n)} \in \ell^\infty$ this shows that $b \in \ell^\infty$. Moreover, we see that for any $m > N$

$$|b_k - a_k^{(m)}| < |b_k - a_k^{(n)}| + |a_k^{(n)} - a_k^{(m)}| \leq \varepsilon$$

which implies $\|b - a^{(m)}\|_\infty \leq \varepsilon$ for $m > N$. □

2. Show that the closed unit ball in ℓ^p is not compact.

Proof. Consider $a^{(n)} = (n^{-1/p}, \dots, n^{-1/p}, 0, 0, \dots)$ where there are n many $n^{-1/p}$ terms. We know that $\|a^{(n)}\|_p = 1$ so $a^{(n)} \in \ell^p$ for every n but it does not converge in ℓ^p since it is not Cauchy in ℓ^p . □

3. Let $\{M_i, d_i\}_{i=1}^n$ be metric spaces. Show that $M = M_1 \times \dots \times M_n$ is a metric space with metric

$$d(x, y) = \left(\sum_{i=1}^n d_i(x_i, y_i)^p \right)^{1/p}$$

Proof. The only important check is the triangle inequality. To see that, given $x, y, z \in M$ define $(a_k) = (d(x_1, y_1), \dots, d(x_n, y_n), 0, \dots)$; $(b_k) = (d(y_1, z_1), \dots, d(y_n, z_n), 0, \dots)$. Then notice that

$$\begin{aligned} d(x, z) &= \left(\sum_{i=1}^n d_i(x_i, z_i)^p \right)^{1/p} \\ &\leq \left(\sum_{i=1}^n d_i(x_i, y_i)^p + d_i(y_i, z_i)^p \right)^{1/p} \\ &= \|a + b\|_p \\ &\leq \|a\|_p + \|b\|_p \\ &= \left(\sum_{i=1}^n d_i(x_i, y_i)^p \right)^{1/p} + \left(\sum_{i=1}^n d_i(y_i, z_i)^p \right)^{1/p} \\ &= d(x, y) + d(y, z) \end{aligned}$$

□

4. Show that the closed unit ball in $C_o([0, 1])$ with norm $\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|$ is not compact.

Proof. Consider the sequence $f_n(x) = x^n \in C([0, 1])$. Obviously, $\|f_n\|_\infty = 1$ and $f_n \rightarrow 0$ point-wise but it does not have a convergent subsequence because $\|f_n - 0\|_\infty = 1$. $\square$

19. (Pugh) Is $\mathbb{Q}$ homeomorphic to $\mathbb{N}$?

Proof. No, $\mathbb{N}$ has isolated points and $\mathbb{Q}$ does not. $\square$

22. (Pugh) If every closed and bounded subset of a metric space M is compact, does it follow that M is complete?

Proof. Yes, given a Cauchy sequence $(a_n) \subset M$, there exists $N \in \mathbb{N}$ such that for every $n > N$, $d(a_N, a_n) < 1$. Then set $r = \max_{i=1}^N (1, d(a_i, a_N)) + 1$. Then $a_m \in \overline{B_r(a_N)}$ for every m . Since this ball is closed and bounded, it is compact so (a_n) has a convergent subsequence that converges to a point $x \in \overline{B_r(a_N)}$. Then (a_n) converges to x as well. Hence M is complete. $\square$

46. (Pugh) Suppose $A \times B$ is compact in $M \times N$ and A, B are non-empty. Then prove that A, B are compact.

Proof. Let (a_n) be a sequence in A and (b_n) be a sequence in B . Then, $(a_n, b_n)_n$ is a sequence in $A \times B$. By compactness, there exists a convergent subsequence (a_{n_k}, b_{n_k}) converging to a point $(a, b) \in A \times B$. Then $(a_{n_k}) \rightarrow a$ in A and $(b_{n_k}) \rightarrow b$ in B . So A, B are compact. $\square$

46. (Pugh) Suppose that A, B are compact, disjoint, non-empty subsets of M . Prove that there exists $a_0 \in A, b_0 \in B$ such that $d(a_0, b_0) \leq d(a, b)$ for every $a \in A, b \in B$.

Proof. Let $f : B \rightarrow \mathbb{R}$ be given by $f(x) = \text{dist}(A, x) := \inf_{y \in A} d(x, y)$. Since B is compact and the distance function is continuous, f attains a minimum at, say, b_0 . Since A is compact, $\text{dist}(A, b_0) = d(a_0, b_0)$ for some $a_0 \in A$. Then, $d(a_0, b_0) \leq d(a, b_0) \leq d(a, b)$ for any $a \in A, b \in B$. $\square$

52. Let (A_n) be a nested decreasing sequence of non-empty closed sets in a metric space M . If M is complete, and $\text{diam } A_n \rightarrow 0$ as $n \rightarrow \infty$, show that $\bigcap A_n$ is just one point.

Proof. Let $a_n \in A_n$. Since the diameters are going to 0, (a_n) is a Cauchy sequence and has a limit $a \in \bigcap A_n$. If there is another point $b \in A_n$, then $0 = \text{diam } A_n \geq |a - b| \implies a = b$. $\square$

Week 3

44. (Pugh) Consider the function $f : M \rightarrow \mathbb{R}$.

- Prove that if f is continuous then its graph is closed.

Proof. Let (x, y) be a limit point of the graph. Then take a sequence (x_n, y_n) converging to this point. By continuity of f , $y = \lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n) = f(x)$ which shows (x, y) is in the graph hence the graph of f is closed. $\square$

- If f is continuous and M is compact, then the graph of f is compact.

Proof. Let (x_n, y_n) be a sequence in the graph. Then, by compactness of M , there exists a convergent subsequence (x_{n_k}) of (x_n) converging to some point $x \in M$. By continuity of f , $(f(x_{n_k}))$ is a convergent subsequence in $\mathbb{R}$, converging to some y . Then $(x_{n_k}, f(x_{n_k}))$ is a convergent subsequence in the graph, converging to the point (x, y) in the graph. $\square$

- If the graph of f is compact, then f is continuous.

Proof. By previous problems, it suffices to prove that $(f(x_k))_k$ converges whenever (x_k) does. Suppose (x_k) converges. Then $(x_k, f(x_k))$ has a convergent subsequence $(x_{n_k}, f(x_{n_k}))$ by the compactness of the graph of f . Let $y = \lim_{k \rightarrow \infty} f(x_{n_k})$ and $x = \lim_{k \rightarrow \infty} x_{n_k}$. Then $y = f(x) = f(\lim_{n \rightarrow \infty} x_n)$ which implies that $\lim_{n \rightarrow \infty} f(x_n) = y$ as well. Hence f is continuous. $\square$

- Show that f is not necessarily continuous if the graph is merely closed.

Proof. Consider $f(x) = \begin{cases} 1 & x = \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$ The graph is closed but f is not continuous. $\square$

54. (Pugh) Suppose $f : S \rightarrow \mathbb{R}$ is a uniformly continuous function defined on a subset S of a metric space M . Prove that f extends uniquely to a uniformly continuous function $\bar{f} : \bar{S} \rightarrow \mathbb{R}$.

Proof. Define $\bar{f}(x) = \lim_{n \rightarrow \infty} f(x_n)$ for $x \in \bar{S} - S$ where (x_n) is any subsequence converging to x . Then, given $\varepsilon > 0$, there exists $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon/2$ for every $x, y \in S$. For any $x \in \bar{S} - S$, if $|x - y| < \delta$, then $|\bar{f}(x) - \bar{f}(y)| = \lim_{n \rightarrow \infty} |f_n(x) - f(y)| \leq \varepsilon/2 < \varepsilon$. So $\bar{f}$ is uniformly continuous over $\bar{S}$. The extension is unique since on limit points $x \in \bar{S} - S$, any two extensions must agree by continuity. $\square$

58. (Pugh) Is the closure of a disconnected set disconnected? What about the interior of a disconnected set?

Proof. No, consider $[0, 1) \cup (1, 2]$. It is disconnected but its closure is connected. For the second question, consider the disconnected set $[0, 1] \cup \{2\}$. Its interior is $(0, 1)$ which is connected. $\square$

59. Prove that every countable metric space is disconnected.

Proof. Let $E = \{a_1, a_2, a_3, \dots\}$ be a countable metric space. Let $r = d(a_1, a_2) > 0$. Then, the set $\{d(a_1, a_k)\}_{k=2}^\infty$ is a countable subset of $\mathbb{R}$ hence there exists a number $s \in (0, r)$ that is not in this set. Then, the sets $B_s(a_1)$ and $B = \{a_j \in E : d(a_j, a_1) > s\}$ are open and $E = B_s(a_1) \cup B$ so E is disconnected. $\square$

96. (Pugh) If $A \subset B \subset C$ and A is dense in B , B is dense in C . Then A is dense in C .

Proof. By definition of dense, $\bar{A} = B, \bar{B} = C$ then B is closed by the first and $\bar{A} = B = \bar{B} = C$ proves A is dense in C . $\square$

101. (Pugh) Let Σ be the set of all infinite sequences of ones and zeroes. Define the metric $d(a, b) = \sum \frac{|a_n - b_n|}{2^n}$ where $a = (a_n), b = (b_n)$ are points in Σ . Prove that Σ is compact and that it is homeomorphic to the Cantor set.

Proof. Let $X = (a^{(n)})_n$ be a sequence in Σ . Either there are infinitely many elements whose first entry is 1 or there are infinitely many elements whose first entry is 0. WLOG, suppose there are infinitely many elements whose first entry is 1. Set $A_1 = \{a^{(m)} \in X : a_1^{(m)} = 1\}$. Then A_1 has infinitely many elements. Similarly, we can assume without loss of generality that there are infinitely many elements in A_1 , whose second entries are 1. Define $A_2 \subset A_1$ to be the set of such elements. Continuing this way, we obtain a nested sequence of sets $\cdots \subset A_3 \subset A_2 \subset A_1$ where the first n entries of elements in A_n are all 1. Now construct a subsequence $(b^{(n)})$ of X by picking $b^{(n)} \in A_n$. Obviously, $b^{(n)}$ converges point-wise to the element $b = (1, 1, 1, 1, \dots) \in \Sigma$. Moreover, given any $\varepsilon > 0$, there exists N such that $\sum_{n>N} \frac{1}{2^n} < \varepsilon$, hence

$$d(b^{(n)}, b) \leq \sum_{j>N} \frac{1}{2^j} < \varepsilon \quad \forall n > N$$

which proves $b^{(n)} \rightarrow b$ in Σ .

Σ is homeomorphic to the cantor set because each element $a \in \Sigma$ can be viewed as a mapping $f(n) : \mathbb{N} \rightarrow \{0, 1\}$ and this map f can be associated to the element of the Cantor set obtained by intersecting the sets that contains the one above. With 1 meaning the right most segment and 0 meaning the right-most segment. This question also shows that C is compact and uncountable. $\square$

115. (Pugh) Rotate the unit circle by a fixed angle α . If α/π is rational, show that each orbit is a finite set. If α/π is irrational show that each orbit is infinite and has closure equal to S^1 .

Proof. If $\alpha/\pi = a/b$ and $\theta \in [0, 2\pi)$, then $\mathcal{O}_\theta = \{\theta + k\alpha : k \in \mathbb{Z}\}$. For $k = 2b$ turns, we arrive at $\theta + 2b\frac{a}{b}\pi = \theta + 2\pi a$ which is the same point as θ on the circle. So the orbit has finitely many elements. If α/π is irrational, then notice that $\theta + k\alpha = \theta + 2\pi l \iff \alpha/\pi = 2l/k \in \mathbb{Q}$ hence the orbits have infinitely many points. Suppose the closure isn't S^1 , then there exists an interval I of the circle that is not contained in the orbit $\mathcal{O}_{(1,0)}$. Suppose this interval has length $\varepsilon > 0$. If there exists numbers $m > n$ such that the distance between the rotations $m\alpha$ and $n\alpha$, which is equal to $2\pi \cdot \frac{(m-n)\alpha}{2\pi} = (m-n)\alpha = d$, less than ε , then we can apply $m\alpha$ clockwise rotation and then $n\alpha$ counterclockwise rotation to move a distance d on the circle. Doing this consecutively will make us fall into the interval I after some time. Hence, the distance between any two number of iterations cannot be less than ε . This means that the points in the orbits are at least ε distance apart but then we can have at most $\frac{1}{\varepsilon} < \infty$ many points in the orbit. $\square$

2. (Pugh) Let $X \subset \mathbb{R}^m$ be compact and $f : X \rightarrow \mathbb{R}$ continuous. Given $\varepsilon > 0$, show that there is a constant M such that for all $x, y \in X$ we have $|f(x) - f(y)| \leq M|x - y| + \varepsilon$.

Proof. Since f continuous and X compact, $f(X) \subset \mathbb{R}^m$ is also compact. Suppose $|f(x)| < N$. f is also uniformly continuous, so there is $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$. Set $M = 2N/\delta$. Then, for $|x - y| \geq \delta$

$$|f(x) - f(y)| < 2N < |x - y|M < |x - y|M + \varepsilon$$

and for $|x - y| < \delta$, uniform continuity gives

$$|f(x) - f(y)| < \varepsilon < M|x - y| + \varepsilon$$

$\square$

5. (Pugh) Let $f : [0, 1]^2 \rightarrow \mathbb{R}$ be continuous. Prove that $g(x) = \max\{f(x, y) : y \in [0, 1]\}$ is continuous.

Proof. Since $[0, 1]^2$ is compact, f is uniformly continuous. Let $\varepsilon > 0, x \in [0, 1]$ be given. Choose $\delta > 0$ such that $|f(a) - f(b)| < \varepsilon$ whenever $|a - b| < \delta$. Then

$$\begin{aligned} |g(x) - g(x')| &= |\max_y f(x, y) - \max_y f(x', y)| \\ &\leq \max_y |f(x, y) - f(x', y)| \\ &= |f(x, y_0) - f(x', y_0)| \quad \text{attained as } [0, 1] \text{ is compact.} \\ &< \varepsilon \end{aligned}$$

whenever $|x - x'| < \delta$. □

11. (Pugh) Let $U \subset \mathbb{R}^m$ be an open set. Suppose that the map $h : U \rightarrow \mathbb{R}^m$ is a homeomorphism onto $\mathbb{R}^m$ that is uniformly continuous. Prove that $U = \mathbb{R}^m$.

Proof. We will show that U is closed. U is already open. Let x be a limit point of U and $(x_n) \rightarrow x, x_n \in U$. Set $y_n = h(x_n)$. By uniform continuity given $\varepsilon > 0, \exists \delta > 0$ such that $|a - b| < \delta \implies |h(a) - h(b)| < \varepsilon$. Then there exists large enough n, m such that $|x_n - x_m| < \delta$ hence $|y_n - y_m| < \varepsilon$ so (y_n) is a Cauchy sequence. Hence there is a limit y . Then, since h is a homeomorphism, h^{-1} is continuous so

$$x = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} h^{-1}(y_n) = h^{-1}(\lim_{n \rightarrow \infty} y_n) = h^{-1}(y)$$

which shows that $x \in U$. Hence $U = \mathbb{R}^m$. □

Week 4

3. (Pugh) Let $f_n : [a, b] \rightarrow \mathbb{R}$ be a sequence of piecewise continuous functions, each of which is continuous at $x_0 \in [a, b]$. Assume $f_n \rightarrow f$ uniformly. Prove that f is continuous at x_0 and disprove that f is piecewise continuous.

Proof. Let $\varepsilon > 0$ be given. Since $f_n \rightarrow f$ uniformly, there exists f_n such that $|f(x) - f_n(x)| < \varepsilon/3$ for every $x \in [a, b]$. Let $\delta > 0$ be such that $|f(x_0) - f(x)| < \varepsilon/3$ whenever $|x - x_0| < \delta$. Then, for $|x - x_0| < \delta$

$$\begin{aligned} |f(x) - f(x_0)| &< |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)| \\ &\leq 3 \cdot \varepsilon/3 = \varepsilon \end{aligned}$$

So f is continuous at x_0 . For the second part, consider the function $f(x) = \begin{cases} \frac{1}{n} & x \in (\frac{1}{n+1}, \frac{1}{n}] \\ 0 & x = 0 \end{cases}$

Then f is not piecewise continuous, because it has infinitely many discontinuities. However, the sequence of functions $f_n = f \cdot \chi_{(\frac{1}{n}, 1]}$ is piecewise continuous and converges uniformly to f . □

4. (Pugh) If $f_n : \mathbb{R} \rightarrow \mathbb{R}$ are uniformly continuous and $f_n \rightarrow f$ uniformly, prove that f is uniformly continuous.

Proof. Let $\varepsilon > 0$ be given. Since convergence is uniform, there exists f_n with $\|f - f_n\|_\infty < \varepsilon/3$. Since f_n is uniformly continuous, there exists $\delta > 0$ such that whenever $|x - y| < \delta$, $|f_n(x) - f_n(y)| < \varepsilon/3$. Then we obtain

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \\ &\leq 3 \cdot \varepsilon/3 = \varepsilon \end{aligned}$$

□

9. (Pugh) If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and the sequence $f_n(x) = f(nx)$ is equicontinuous, what can be said about f ?

Proof. f is constant in this case. Suppose $f(x) \neq f(y)$ for distinct $x, y \in \mathbb{R}$. Then, let $r = |f(x) - f(y)| > 0$ and set $\varepsilon = r/2$. By equicontinuity, there exists $\delta > 0$ such that $|a - b| < \delta \implies |f_n(a) - f_n(b)| < \varepsilon$ for all n . Then, there exists n large enough so that $\frac{|x-y|}{n} < \delta$. However now

$$r = |f(x) - f(y)| = |f_n(x/n) - f_n(y/n)| < \varepsilon = r/2 \implies r < r/2$$

is a contradiction. Hence $f(x)$ is constant. □

12. (Pugh) Prove that if $\mathcal{E} \subset C_0(M, N)$ is equicontinuous then so is its closure.

Proof. Let $\varepsilon > 0$ be given. Then there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon/3$ whenever $|x - y| < \delta$ for every $f \in \mathcal{E}$. Let $g \in \overline{\mathcal{E}} - \mathcal{E}$ and let (g_n) be a sequence of functions converging to g uniformly. Then there exists g_n such that $\|g - g_n\|_\infty < \varepsilon/3$. Then we have

$$\begin{aligned} |g(x) - g(y)| &\leq |g(x) - g_n(x)| + |g_n(x) - g_n(y)| + |g_n(y) - g(y)| \\ &< \varepsilon \end{aligned}$$

whenever $|x - y| < \varepsilon$. Hence $\overline{\mathcal{E}}$ is equicontinuous as well. □

13. (Pugh) Suppose that $f_n : \mathbb{R} \rightarrow \mathbb{R}$ is a sequence of continuous functions and for each compact subset $K \subset \mathbb{R}$, the restricted sequence $(f_n|_K)$ is pointwise bounded and pointwise equicontinuous.

a) Does it follow that there is a subsequence of (f_n) that converges point-wise to a continuous limit function $\mathbb{R} \rightarrow \mathbb{R}$?

b) What about uniform convergence?

Proof. a) Yes. Let $x_0 \in \mathbb{R}$. $(f_n(x_0))_n$ is a bounded sequence so $s = \sup_n f_n(x_0)$ exists. Let $f_{n_k}(x_0) \rightarrow s$. By point-wise equicontinuity there exists $\delta > 0$ such that $|x - x_0| < \delta \implies |f_n(x) - f_n(x_0)| < \varepsilon$. Then for large k, ℓ one has

$$|f_{n_\ell}(y) - f_{n_k}(y)| \leq |f_{n_\ell}(y) - f_{n_\ell}(x)| + |f_{n_\ell}(x) - f_{n_k}(x)| + |f_{n_k}(x) - f_{n_k}(y)| < 3\varepsilon$$

Hence $(f_{n_k})_k$ converges on every point in $(x_0 - \delta, x_0 + \delta)$. By continuity of each f_{n_k} , this subsequence also convergence on the closure $[x_0 - \delta, x_0 + \delta]$. Now, let $K = [-m, m]$. By compactness of K , we can start at $x_0 = -m$ and cover K with finitely many open intervals $A_i = (x_i - \delta_i, x_i + \delta_i)$ such that $x_i = x_{i-1} + \delta_{i-1}$ for $i \geq 1$ and $(f_{n_k})_k$ converges on each A_i by the above argument. Then f_{n_k} converges on $[-m, m]$ for any m so f_{n_k} converges on $\mathbb{R}$.

Uniform convergence is not always true. Consider $f_n(x) = \begin{cases} x^n & |x| < 1 \\ \frac{1}{x^n} & |x| > 1 \end{cases}$. This sequence converges point-wise to $f(x) = \begin{cases} 1 & x = 1 \\ 0 & x \neq 1 \end{cases}$. The sequence is pointwise equicontinuous and pointwise bounded over compact subsets of $\mathbb{R}$. $\square$

21. (Pugh) Suppose $\mathcal{E} \subset C^0$ is equicontinuous and bounded. Prove that $g(x) := \sup\{f(x) : f \in \mathcal{E}\}$ is a continuous function of x . Show that this claim is false without the equicontinuity assumption.

Proof. Let $\varepsilon > 0, x \in \mathbb{R}$. By equicontinuity, there is $\delta > 0$ such that $|a-b| < \delta \implies |f(a)-f(b)| < \varepsilon$. Then for $|x-y| < \delta$,

$$|g(x) - g(y)| = \left| \sup_f f(x) - \sup_f f(y) \right| \leq \sup_f |f(x) - f(y)| < \varepsilon$$

so g is continuous at x . In fact, g is uniformly continuous. Without the equicontinuity of $\mathcal{E}$, we can take $f_n(x) = 1 - (1-x)^n$. Then f_n is not a family of equicontinuous functions and $g(0) = 0, g(x) = 1$ for any $x \neq 0$. So g is not continuous. $\square$

2. (Pugh) Suppose that $f_n \in C^0$ and for each $x \in [a, b]$,

$$f_1(x) \geq f_2(x) \geq \dots$$

and $\lim_{n \rightarrow \infty} f_n(x) = 0$. Is this sequence equicontinuous?

Proof. Yes. Notice that since $|f_n|$ is continuous over a compact set, it attains its maximum. Hence, $\|f_n\|_\infty = |f_n(x_n)|$ for some $x_n \in [a, b]$. Moreover, since $f_n(x) > f_{n+1}(x)$ and $f_n(x) \xrightarrow{n \rightarrow \infty} 0$, we must have $f_n(x) > 0$ for every n, x . Then we see that $\|f_n\|_\infty = f_n(x_n)$ is a decreasing sequence since $f_{n+1}(x_{n+1}) \leq f_n(x_{n+1}) \leq f_n(x_n)$. Now, the sequence $(x_n)_n$ is bounded, hence there exists a converging subsequence $(x_{n_k})_k$ converging to some $x^* \in [a, b]$. Notice that given $\varepsilon > 0$ there exists n such that $|f_n(x^*)| < \varepsilon$ then for large enough ℓ , we have $n_\ell > n$ and

$$\lim_{k \rightarrow \infty} f_{n_k}(x_{n_k}) \leq \lim_{k \rightarrow \infty} f_{n_\ell}(x_{n_k}) = f_{n_\ell}(x^*) < \varepsilon$$

which shows $\lim_{k \rightarrow \infty} f_{n_k}(x_{n_k}) = 0$. Since $(f_n(x_n))_n$ is a sequence of decreasing numbers bounded below by 0, the existence of a subsequence $(f_{n_k}(x_{n_k}))_k$ that converges to 0 implies that $\lim_{n \rightarrow \infty} f_n(x_n) = 0$. This shows that $f_n \rightarrow 0$ uniformly since given $\varepsilon > 0$, $f_n(x_n) < \varepsilon$ for large enough n and $f_n(x) \leq f_n(x_n) < \varepsilon$.

Finally, to prove that (f_n) is equicontinuous, let $\varepsilon > 0$ be given. There exists N such that $\|f_n\|_\infty < \varepsilon/2$ for $n > N$. Then $|f_n(x) - f_n(y)| < \varepsilon$ for $n > N$. Each f_m is uniformly continuous, being continuous over a compact subset. So let δ_i be the corresponding bound for f_i for $i = 1, 2, \dots, N$. Let $\delta = \min_i \delta_i$. Then whenever $|x - y| < \delta$, $|f_n(x) - f_n(y)| < \varepsilon$ for any $n \in \mathbb{N}$. $\square$

Week 5

1. Let X be a compact metric space and $C_0(X)$ be the space of continuous bounded functions with metric $D(f, g) = \sup_{x \in X} |f(x) - g(x)|$. If $\mathcal{F} \subset C_0(X)$ is point-wise equicontinuous, then prove that $\mathcal{F}$ is equicontinuous.

Proof. For any $x \in X$, there exists $\delta_x > 0$ such that $|x - y| < \delta_x \implies |f(x) - f(y)| < \varepsilon$ for all $f \in \mathcal{F}$. Since X is compact, $X \subset B_{\delta_1}(x_1) \cup B_{\delta_2}(x_2) \cup \dots \cup B_{\delta_n}(x_n)$. Put $\delta = \min_i \delta_i$ to obtain equicontinuity. $\square$

2. Starting from $(a - b)^2 \geq 0$ prove the Cauchy-Schwartz inequality

$$\sum_{n=1}^{\infty} |a_n b_n| \leq \left(\sum_{n=1}^{\infty} |a_n|^2 \right)^{1/2} \cdot \left(\sum_{n=1}^{\infty} |b_n|^2 \right)^{1/2}$$

and show that Cauchy-Schwartz is a special case of Holder's inequality.

Proof. Define $f(x) = \sum_{n=1}^N (a_n x - b_n)^2 = (\sum_{n=1}^N |a_n|^2) x^2 - 2(\sum_{n=1}^N a_n b_n) x + (\sum_{n=1}^N |b_n|^2)$. Using the fact that $f(x) \geq 0$, we must have $\Delta \leq 0$ letting $N \rightarrow \infty$. $\square$

3. Let $(V, \|\cdot\|)$ be a normed vector space. Show that $v \rightarrow \|v\|$ is continuous.

Proof. Basically, $||v| - |w|| \leq \|v - w\|$. $\square$

4. Let $A \subset \ell^\infty$ be the set

$$A = \{(a_n)_n : \lim_{n \rightarrow \infty} a_n = 0\}$$

Show that A is a closed, linear subspace of ℓ^∞ . Is A compact? Is $A \cap \{a \in \ell^\infty : \|a\|_\infty \leq 1\}$ compact?

Proof. Let x be a limit point of A and let $x^{(n)}$ be a sequence converging to it. Let $\varepsilon > 0$ be given, then there exists $x^{(n)}$ such that $\|x - x^{(n)}\|_\infty < \varepsilon/2$. And there exists M such that for all $m > M$, $|x_m^{(n)}| < \varepsilon/2$. Therefore, for $m > M$,

$$|x_m| < |x_m - x_m^{(n)}| + |x_m^{(n)}| < \|x - x^{(n)}\|_\infty + |x_m^{(n)}| < \varepsilon \implies \lim_{m \rightarrow \infty} x_m = 0 \implies x_m \in A$$

Hence A is closed. It is also linear, obviously. It is not compact, consider the sequence $x^{(n)} = (1, \dots, 1, 0, \dots)$ where there are n many 1s. Then $x^{(n)} \xrightarrow{n \rightarrow \infty} (1, 1, 1, 1, \dots) \notin A$ point-wise, so there does not exist a converging subsequence. Hence A is not compact. $\square$

5. Let $f : M \rightarrow N$ be a uniformly continuous map. Show that if M is totally bounded, then $f(M)$ is totally bounded.

Proof. Let $\varepsilon > 0$ be given. There exists $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$. Then $M \subset B_\delta(x_1) \cup \dots \cup B_\delta(x_n)$. Then $\bigcup_i B_\varepsilon(f(x_i)) \supset N$ since $f(B_\delta(x_i)) \subset B_\varepsilon(f(x_i))$. $\square$

6. Let $\mathcal{F} \in \ell^1$ be a closed set with the following properties.
 (i) For all $\varepsilon > 0$, there is an $N > 0$ such that

$$\sum_{j \geq N} |x_j| < \varepsilon$$

for all $x \in \mathcal{F}$.

- (ii) For all $x \in \mathcal{F}$, we have $\|x\|_1 \leq 1$.

Show that $\mathcal{F}$ is compact.

Proof. Let $(x^{(n)})_n \subset \mathcal{F}$ be a sequence in $\mathcal{F}$. We want to show there exists a converging subsequence. Since $\{x_1^{(n)}\}_n$ is bounded in $\mathbb{R}$, it has a convergent subsequence so we can extract a subsequence $(x^{(1,k)})_k \subset (x^{(n)})_n$ such that $\lim_{k \rightarrow \infty} x_1^{(1,k)}$ exists. Similarly, we can extract a subsequence $(x^{(2,k)})_k \subset (x^{(1,k)})_k$ such that $\lim_{k \rightarrow \infty} x_i^{(2,k)}$ exists for $i = 1, 2$. We keep extracting subsequences this way and then consider the diagonal sequence $(x^{(n,n)})_n$. By construction, $\lim_{k \rightarrow \infty} x_m^{(k,k)}$ exists for any $m \in \mathbb{N}$. We show that this sequence is Cauchy. Then by the completeness of ℓ^1 it converges to a point $x \in \ell^1$. This element $x = (x_n)$ must be $x_n = \lim_{k \rightarrow \infty} x_n^{(k,k)}$ by the point-wise convergence of the entries. By the continuity of the absolute value function, $\sum_{j \geq N} |x_j| < \varepsilon$ and $\|x\|_1 \leq 1$ so $x \in \mathcal{F}$. Now let's prove that $(x^{(n,n)})_n$ is Cauchy. Given $\varepsilon > 0$, there is N such that

$$\begin{aligned} \|x^{(n,n)} - x^{(m,m)}\|_1 &\leq \sum_{j=1}^{N-1} |x_j^{(n,n)} - x_j^{(m,m)}| + \sum_{j \geq N} |x_j^{(n,n)}| + \sum_{j \geq N} |x_j^{(m,m)}| \\ &\leq \sum_{j=1}^{N-1} |x_j^{(n,n)} - x_j^{(m,m)}| + 2\varepsilon \end{aligned}$$

For $n > m > N$, $x^{(n,n)}, x^{(m,m)} \in (x^{(N,k)})_k$ and for each $j \in \{1, \dots, N-1\}$, there exists k_j such that whenever $k, \ell > k_j$,

$$|x_j^{(N,k)} - x_j^{(N,\ell)}| < \frac{\varepsilon}{N-1}$$

Set $k = \max_j k_j$. Then for $n, m > \max(k, N)$, we have

$$\sum_{j=1}^{N-1} |x_j^{(n,n)} - x_j^{(m,m)}| < \varepsilon$$

so

$$\|x^{(n,n)} - x^{(m,m)}\|_1 < 3\varepsilon$$

for large enough n, m . □

7. Let $f_n \in C_0(X)$ be a sequence of uniformly continuous functions that converge uniformly to some f . Show that f_n are equicontinuous.

Proof. Let $\varepsilon > 0$. Then there exists N such that $\|f_n - f\|_\infty < \varepsilon/3$ for every $n > N$. There also exists $\delta_0 > 0$ such that $|x - y| \leq \delta_0 \implies |f(x) - f(y)| < \varepsilon/3$ since the uniform limit of uniformly continuous functions is uniformly continuous. Then, there also exists $\delta_i > 0$ such that

$|x - y| < \delta_i \implies |f_i(x) - f_i(y)| < \varepsilon$ for every $i = 1, 2, \dots, N$. Set $\delta = \min_i \delta_i, \delta_0$. Then given $|x - y| < \delta$, $|f_i(x) - f_i(y)| < \varepsilon$ for $i = 1, \dots, N$ and for $n > N$,

$$\begin{aligned} |f_n(x) - f_n(y)| &\leq |f_n(x) - f(x)| + |f(x) - f(y)| + |f(y) - f_n(y)| \\ &\leq \varepsilon \end{aligned}$$

Hence $(f_n)_n$ is equicontinuous. $\square$

8. Let (M, d) be a metric space and $E \subset M$ be a non-empty subset. Define

$$\text{dist}_E(x) := \inf_{y \in E} d(x, y)$$

Show that $\text{dist}_E(x)$ is a continuous function from M to $\mathbb{R}$ and is 0 iff $x \in \overline{E}$.

Proof. Let $x \in M$ and $\varepsilon > 0$ be given. Since $|d(y, x) - d(y, z)| \leq d(x, z)$, we have

$$\left| \inf_{y \in E} d(y, x) - \inf_{y \in E} d(y, z) \right| \leq \inf_{y \in E} |d(y, x) - d(y, z)| \leq d(x, z)$$

hence dist_E is a continuous function. Moreover, $\text{dist}_E(x) = 0 \iff \forall \varepsilon > 0, \exists y \in E \cap B_\varepsilon(x) \iff x \in \overline{E}$. $\square$

Week 6

25. (Pugh) Suppose $f : M \rightarrow M$ is a contraction, but M is not necessarily complete. Prove that f is uniformly continuous. Why does this fact imply that f extends uniquely to a continuous map $\widehat{f} : \widehat{M} \rightarrow \widehat{M}$ where $\widehat{M}$ is the completion of M ? Is $\widehat{f}$ a contraction?

Proof. Let $\varepsilon > 0$ be given. There exists $\alpha \in (0, 1)$.

$$d(f(x), f(y)) < \alpha \cdot d(x, y)$$

Choose $\delta = \varepsilon/\alpha$. Then obviously $d(f(x), f(y)) < \varepsilon$ whenever $d(x, y) < \delta$. Let $\{x_n\}$ be a Cauchy sequence in M , converging to $\hat{x} \in \widehat{M}$. Then by continuity of f , $\{f(x_n)\}$ is a Cauchy sequence too. Say it converges to some $x' \in \widehat{M}$. Then extend f by setting $\widehat{f}(\hat{x}) = x'$. This is a continuous map because given $\varepsilon > 0$, and $x', y' \in \widehat{M}$, let $x_n \rightarrow x', y_n \rightarrow y'$. By definition there exists $\delta > 0$ for the uniform continuity of f . Then if $|x' - y'| < \delta/4$ there exists $x_n \in B_{\delta/4}(x'), y_m \in B_{\delta/4}(y')$ such that $|\widehat{f}(x') - f(x_n)| < \varepsilon, |\widehat{f}(y') - f(y_m)| < \varepsilon$ so

$$|\widehat{f}(x') - \widehat{f}(y')| < |\widehat{f}(x') - f(x_n)| + |f(x_n) - f(y_m)| + |f(y_m) - \widehat{f}(y')| < 3\varepsilon$$

the middle distance is $< \varepsilon$ because $|x_n - y_m| < \delta$. This extension is now unique by continuity. The extension is a contraction because we can take the limit on the contraction inequality and get $d(f(x'), f(y')) < \alpha \cdot d(x', y')$ whenever $x', y' \in \widehat{M}$. $\square$

26. (Pugh) Given an example of a contraction of an incomplete metric space that has no fixed point.

Proof. Take $f(x) = x/3$. Then $\alpha = 2/3$ is a contraction constant. Let the space be $M = \mathbb{R} \setminus \{0\}$. $\square$

5. (Pugh) Let (a_n) be a sequence of non-zero real numbers. Prove that the sequence of functions

$$f_n(x) = \frac{1}{a_n} \sin(a_n x) + \cos(x + a_n)$$

has a subsequence converging to a continuous function.

Arzela-Ascoli. We need to check that this family of functions are uniformly bounded and equicontinuous. Firstly, if $\liminf_n |a_n| > 0$ then there exists $M \in \mathbb{R}$ such that all but finitely many of the f_n satisfy

$$\|f_n\|_\infty < \frac{1}{M} + 1$$

Hence f_n are uniformly bounded. (To find the uniform bound, take the maximum among $\frac{1}{M} + 1$, and the finitely many of the $\|f_j\|_\infty$.) If $\liminf_n |a_n| = 0$ then there exists a subsequence $\{a_{n_k}\}$ such that $\lim_{k \rightarrow \infty} |a_{n_k}| = 0$ then we have

$$\lim_{k \rightarrow \infty} \frac{1}{a_{n_k}} \cdot \sin(a_{n_k} \cdot x) = 1$$

so $|1 - \frac{1}{a_{n_k}} \cdot \sin(a_{n_k} \cdot x)| < 0.10$ for all but finitely many k . Then, we can bound all but finitely many of the f_{n_k} by

$$\|f_{n_k}\|_\infty < 2.10$$

for $\{f_n\} \setminus \{f_{n_k}\}$, the bound from previous part applies. Hence we uniformly bound all f_n this way. Finally, to see that f_n are equicontinuous, notice that $\sin$ and $\cos$ are Lipschitz continuous since

$$\frac{|\sin x - \sin y|}{|x - y|} = |\sin'(t)| = |\cos t| \leq 1 \implies |\sin x - \sin y| \leq |x - y|$$

and similarly for $\cos$. Hence, given ε , for any $|x - y| \leq \varepsilon/2$ we have

$$\begin{aligned} |f_n(x) - f_n(y)| &\leq \frac{1}{|a_n|} \cdot |\sin(a_n x) - \sin(a_n y)| + |\cos(a_n + x) - \cos(a_n + y)| \\ &\leq \frac{|a_n|}{|a_n|} |x - y| + |x - y| \\ &\leq 2|x - y| \\ &\leq \varepsilon \end{aligned}$$

Then by Arzela-Ascoli, there exists a convergent subsequence, converging uniformly to some function f . $\square$

26. (Pugh) Let f_n be a sequence of continuous maps $[0, 1] \rightarrow \mathbb{R}$ such that

$$\int_0^1 f_n(y)^2 dy \leq 5$$

for all n . Define $g_n : [0, 1] \rightarrow \mathbb{R}$ by

$$g_n(x) = \int_0^1 \sqrt{x+y} f_n(y) dy$$

Find a constant $K \geq 0$ such that $|g_n(x)| \leq K$ for all n and prove that a subsequence of g_n converges uniformly.

Proof. By Cauchy-Schwartz,

$$|g_n(x)| \leq \int_0^1 |\sqrt{x+y} f_n(y)| dy \leq \left(\int_0^1 (x+y) dy \right)^{1/2} \cdot \left(\int_0^1 |f_n(y)|^2 dy \right)^{1/2} \leq \sqrt{10}$$

so $K = \sqrt{10}$. This family is also equicontinuous. Let $\varepsilon > 0$ be given. Then for $|x - y| \leq \varepsilon^2/\sqrt{5}$ we have

$$\begin{aligned} |g_n(x) - g_n(y)| &= \left| \int_0^1 \sqrt{x+t} f_n(t) dt - \int_0^1 \sqrt{y+t} f_n(t) dt \right| \\ &\leq \int_0^1 |f_n(t) \cdot (\sqrt{x+t} - \sqrt{y+t})| dt \\ &\leq \left(\int_0^1 |f_n(t)|^2 dt \right)^{1/2} \cdot \left(\int_0^1 |\sqrt{x+t} - \sqrt{y+t}|^2 dt \right)^{1/2} \\ &\leq \sqrt{5} \cdot \left(\int_0^1 |x-y| dt \right)^{1/2} \\ &= \sqrt{5} \cdot \sqrt{|x-y|} \\ &\leq \varepsilon \end{aligned}$$

By Arzela-Ascoli there exists a converging subsequence. □

39. (Pugh) Let $f : [0, \pi] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^\pi f(x) \sin(nx) dx = 0$$

for all integers $n \geq 1$. Prove that f is identically constant.

Proof. Trigonometric polynomials are dense in 2π -periodic functions. So let $\{p_n\}$ be a sequence of trigonometric polynomials that approximate f .

$$p_n(x) = a_0^n + \sum_{i=1}^{n_k} a_i \cos(ix) + b_i \sin(ix)$$

Then

$$\int_0^\pi p_n(x) \sin(mx) dx = a_0^n \cdot \frac{\pi}{2} + \sum_{2 \nmid k-m} a_k^n \cdot \left(\frac{1}{k-m} + \frac{1}{k+m} \right) + b_m^n \cdot \frac{\pi}{2}$$

Sending $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} \left(a_0^n \cdot \frac{\pi}{2} + \sum_{2 \nmid k-m} a_k^n \cdot \left(\frac{1}{k-m} + \frac{1}{k+m} \right) + b_m^n \cdot \frac{\pi}{2} \right) = \int_0^\pi f(x) \sin(mx) dx = 0$$

if and only if $\lim_{n \rightarrow \infty} a_m^n, b_m^n, a^n = 0$ hence $f = 0$. □

41. Let $f : [0, \infty) \rightarrow [0, \infty)$ be a monotonically decreasing function with

$$\int_0^\infty f(x) dx < \infty$$

Prove that $\lim_{x \rightarrow \infty} xf(x) = 0$.

Proof. Since x decreases monotonically,

$$\frac{x}{2}f(x) \leq \int_{x/2}^x f(y) dy \leq \int_{x/2}^\infty f(y) dy \xrightarrow{x \rightarrow \infty} 0 \quad (1)$$

□

60. (Pugh) Does there exist a continuous real-valued function $f(x)$, $0 \leq x \leq 1$, such that

$$\int_0^1 xf(x) dx = 1$$

and

$$\int_0^1 x^n f(x) dx = 0$$

for all $n = 0, 2, 3, 4, \dots$?

Proof. Suppose such a function f exists then it follows that $\langle f, x^n \rangle = 0$ for every $n \neq 1$ where the inner product of two functions is given by $\langle f, g \rangle = \int_0^1 fg$. Let M be the set of polynomials without terms with x . Then M is an algebra as it is closed under scalar multiplication, addition and multiplication of the polynomials. M is nowhere vanishing (constant polynomials) and it separates points ($x^3 \in M$). Hence by the Stone-Weierstrass theorem M is dense in $C_0([0, 1])$. Then x can be approximated by polynomials in M . So $\langle f, x \rangle = 0$ as well. □

1. Let R be the set of Riemann integrable functions $f : [0, 1] \rightarrow \mathbb{R}$. For $p \in [1, \infty)$ let

$$A_p := \{f \in R : \int_0^1 |f(s)|^p ds \leq 1\}$$

For any $f \in A_p$, define $F : [0, 1] \rightarrow \mathbb{R}$ by

$$F(t) = \int_0^t f(s) ds$$

For which values of $p \in [1, \infty)$, is the set of such F equicontinuous?

Proof. If $p > 1$, for $|x - y| \leq \varepsilon^{p/(p-1)}$ and for any $f \in A_p$

$$\begin{aligned} |F(x) - F(y)| &\leq \int_y^x |f(t)| dt \\ &\leq \left(\int_y^x |f(t)|^p dt \right)^{1/p} \cdot \left(\int_y^x 1 dt \right)^{1-1/p} && \text{Holder's} \\ &\leq |x - y|^{1-1/p} \\ &< \varepsilon \end{aligned}$$

So such F is equicontinuous. If $p = 1$, consider $f_n(x) = x^{n-1} \cdot n$ then the corresponding $F(x) = x^n$ so it is not equicontinuous but $f_n \in A_1$. $\square$

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Show that there exists a sequence of polynomials $p_n(x)$ such that $\lim_{n \rightarrow \infty} p_n(x) = f(x)$ for all $x \in \mathbb{R}$. Find all bounded continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ such that there is a sequence of polynomials $p_n(x)$ converging uniformly to f on $\mathbb{R}$.

Proof. Since the set of polynomials are dense in $C_0([a, b])$, for any $n \in \mathbb{N}$, there exists a polynomial $p_n(x)$ such that

$$\|f - p_n\| \leq \frac{1}{n}$$

on $[-n, n]$. Then the sequence $(p_n)_n$ converge to f point-wise on all of $\mathbb{R}$. To see this, let $\varepsilon > 0$ be given. Then for n large enough, say $n > N$, one has $\frac{1}{n} < \varepsilon$ and $x \in [-n, n]$ so

$$|f(x) - p_n(x)| \leq \|f - p_n\|_\infty \leq \frac{1}{n} < \varepsilon, \quad \forall n > N$$

For the second part of the question, suppose there is a sequence $(p_n)_n$ converging uniformly to some f . If p_n is non-constant then $\lim_{x \rightarrow \infty} p_n(x) = \pm\infty$. Since f is bounded, this shows that only finitely many p_n must be non-constant. Then we can remove those from this sequence and still approximate f with the remaining polynomials. The remaining polynomials are all constant hence f must be a constant function. $\square$

3. Let $F_n : \mathbb{R} \rightarrow \mathbb{R}$ be a sequence of continuous functions such that $|F_n(x)| \leq 1$ for all $x \in \mathbb{R}$. Let $u_n : \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$\frac{d}{dt}u_n(t) = F_n(u_n(t)), \quad u_n(0) = 0$$

a) Show that there exists a subsequence of the u_n 's that converge uniformly on compact sets to some u .

b) Suppose $F_n \rightarrow F$ uniformly on compact sets. Show that the subsequence above converges to a solution of

$$u'(t) = F(u(t)) \quad u(0) = 0$$

Proof. To show that u_n is uniformly bounded on compact sets it suffices to show that u_n is uniformly bounded on $[-n, n]$ for any n . So let $x \in [-n, n]$ for some $n \in \mathbb{N}$.

$$\begin{aligned} |u_m(x)| &= \left| u_m(0) + \int_0^x u'_m(t) dt \right| \\ &\leq |u_m(0)| + \int_0^x |u'_m(t)| dt \\ &\leq \int_0^x |F_m(u_m(t))| dt \\ &\leq n \end{aligned}$$

for any u_m . Next up, notice that on $[-n, n]$,

$$|u_m(x) - u_m(y)| < |u'_m(t)| \cdot |x - y| = |F_m(u_m(t))| \cdot |x - y| \leq |x - y|$$

so u_m are equicontinuous. Hence, there exists a convergent subsequence by Arzela-Ascoli. For part b, let u_n converge to a limit function u on $[-m, m]$. Then,

$$|u(t) - \int_0^t F(u(s)) ds| \leq |u(t) - u_n(t)| + \left| u_n(t) - \int_0^t F_n(u_n(s)) ds \right| + \left| \int_0^t F_n(u_n(s)) ds - \int_0^t F_n(u(s)) ds \right| + \left| \int_0^t F_n(u(s)) ds - \int_0^t F(u(s)) ds \right|$$

$|u(t) - u_n(t)| < \varepsilon$ for large enough n . The second absolute value is equal to 0 by definition. We can bound the third one by

$$\left| \int_0^t F_n(u_n(s)) ds - \int_0^t F_n(u(s)) ds \right| \leq \int_0^t |F_n(u_n(s)) - F_n(u(s))| ds$$

since $u_n \rightarrow u$ uniformly and F_n is continuous, $F_n \circ u_n \rightarrow F_n \circ u$ uniformly as well. So for large enough n ,

$$\int_0^t |F_n(u_n(s)) - F_n(u(s))| ds \leq |t| \cdot \varepsilon < n\varepsilon$$

For the fourth term, since $F_n \rightarrow F$ uniformly, for large enough n ,

$$\left| \int_0^t F_n(u(s)) ds - \int_0^t F(u(s)) ds \right| \leq \int_0^t |F_n(u(s)) - F(u(s))| ds \leq n\varepsilon$$

This proves part b. □

4. Let $I_n(x) = \frac{n}{2} \chi_{\{|x| < \frac{1}{n}\}}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a compactly supported function that is continuous for all $x \neq 0$. Assume that the limits $\lim_{x \rightarrow 0^-} f(x)$, $\lim_{x \rightarrow 0^+} f(x)$ exist. Compute $\lim_{n \rightarrow \infty} (f \star I_n)(x)$ for all $x \in \mathbb{R}$.

Proof. Simply compute

$$\begin{aligned} (f \star I_n)(x) &= \int_{\mathbb{R}} f(x-t) I_n(t) dt \\ &= \frac{n}{2} \int_{-1/n}^{1/n} f(x-t) dt \end{aligned}$$

As $n \rightarrow \infty$, apply L'hopitals on $\frac{\int_{-1/n}^{1/n} f(x-t) dt}{2/n}$ to obtain $f(x)$. If $x = 0$, the same procedure gives $\frac{1}{2}(\lim_{x \rightarrow 0^-} f(x) + \lim_{x \rightarrow 0^+} f(x))$. □

Week 7

Definition: Let M be a metric space. A set $S \subseteq M$ is called *somewhere dense* if there is an open set U such that the closure of $S \cap U$ contains U . S is called *nowhere dense* if it is not somewhere dense.

1. Let M be a nonempty complete metric space. If $\{F_n\}$ is a sequence of closed sets each with empty interior, then show that

$$M \neq \bigcup_{n=1}^{\infty} F_n.$$

Proof. For the sake of contradiction suppose M is equal to the union. Then F_n^c is open. We claim it is also dense. Let $x \in M \setminus F_n^c$. Then $x \in F_n$. Since F_n has empty interior, there exists $y_m \in B_{1/m}(x) \cap F_n$ for every m , otherwise $B_{1/m}(x) \subset F_n$ for some m which implies that x is an interior point of F_n . Then x is a limit point of F_n^c so $x \in \overline{F_n^c}$. This proves F_n is dense in M . By Baire Category theorem, $\bigcap_n F_n^c = M^c$ is dense in M but this set is empty! $\square$

2. Let M be a nonempty complete metric space. If F_n is a sequence of closed nowhere dense sets then prove that

$$M \neq \bigcup_{n=1}^{\infty} F_n.$$

Proof. By a similar idea, it suffices to show that F_n^c is dense and open. It is open since F_n is closed. Suppose F_n^c is not dense. Then there exists $x \in M \setminus \overline{F_n^c}$ which implies that $x \in \text{int } F_n$. Then there exists a neighbourhood $U \subset \text{int } F_n$ of x . However, then $\text{cl}(F_n \cap U) = \text{cl}(U)$ so F_n is somewhere dense. This is a contradiction. Hence $M \setminus \overline{F_n^c}$ is empty so F_n^c is dense in M . $\square$

3. Let M be a non-empty, perfect, complete metric space. $E \subset M$ be a countable subset. Show that E is not a countable intersection of open and dense sets.

Proof. A set $S \subset M$ is called meager if it is the countable union of nowhere dense sets. It is called comeager if its complement is meager. With this definition in mind, if $G = \bigcap_n G_n$ where each G_n is open and dense, then each G_n^c is nowhere dense. Hence G is comeager. If E was the countable intersection of open and dense sets, then by Baire category theorem E is dense and comeager. On the other hand, since M is perfect, every singleton $\{x_n\} \subset E$ is a no-where dense set. The closure of any $\{x_n\} \cap S$ is $\{x_n\}$ and since M is perfect, $\overline{S}$ contains more than one point so it cannot be equal to $\{x_n\}$. Therefore E is a meager set as well. But E cannot be meager and comeager at the same time. This is a contradiction. $\square$

4. Let $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}$ be a continuous function such that $\lim_{n \rightarrow \infty} f(nx) = 0$ for every $x \in \mathbb{R}$. Prove that $\lim_{x \rightarrow \infty} f(x) = 0$.

Proof. Let $\varepsilon > 0$ be given and set $E_k = \{x : |f(nx)| \leq \varepsilon, \forall n \geq k\}$. We claim that E_k is closed. Indeed, if x is a limit point of E_k , then there is a sequence $\{x_n\} \rightarrow x$ in E_k then by continuity of f , $|f(nx)| = \lim_{k \rightarrow \infty} |f(nx_k)| \leq \varepsilon \implies x \in E_k$. Next, notice that $\mathbb{R}_{\geq} = \bigcup_k E_k$. Given $x \in \mathbb{R}_{\geq}$, there exist N such that for all $n \geq N$, $|f(nx)| \leq \varepsilon$ since $\lim_{n \rightarrow \infty} f(nx) = 0$ then $x \in E_N$. Now, E_k^c are all open. If they are also dense then by Baire category theorem $\bigcap E_n^c = \{\}$ is dense in $\mathbb{R}$, which is not true. Hence, there exists E_k whose complement is not dense. This shows that E_k contains an open interval (a, b) . Then we claim that $\bigcup_{n \geq k} (na, nb) \supseteq (T, \infty)$ for some T . Since $nb \rightarrow \infty$ as $n \rightarrow \infty$, we need to make sure the lower bound is covered by the union. If for some $n \geq k$, $na > (n-1)b$ then T cannot be in $((n-1)b, na)$. However, since $b > a$, and $\frac{n}{n-1} \xrightarrow{n \rightarrow \infty} 1$, for big enough n , $\frac{n}{n-1} < \frac{b}{a}$ then we can let $T = an$ for one such n . Then we claim that $\limsup_{t \rightarrow \infty} |f(t)| < \varepsilon$. Since for large x , we have $(ma, mb) \ni x$ for some $m \geq k$ and since $\frac{x}{m} \in E_k$

$$|f(x)| = |f(m \cdot \frac{x}{m})| \leq \varepsilon$$

which proves $\lim_{t \rightarrow \infty} f(t) = 0$. $\square$

5. For $v \in \mathbb{R}$, consider the equation

$$f(x, y) = v + \int_0^x \int_0^y st f(s, t)^2 ds dt$$

for a continuous function $f : [0, 1]^2 \rightarrow \mathbb{R}$. Prove that there is an $r > 0$ such that for all v with $|v| < r$ there exists a continuous $f : [0, 1]^2 \rightarrow \mathbb{R}$ that satisfies the above equation.

Proof. Let $M = C_0([0, 1]^2)$. We want to apply the Banach Fixed point theorem. We want to find a closed ball $A \subset M$ of radius R and define $g : A \rightarrow A$ by

$$g(f)(x, y) = v + \int_0^x \int_0^y st f(s, t)^2 ds dt$$

The choice of R should enable us to conclude that g is a contraction.

$$\begin{aligned} \|g(f) - g(h)\|_\infty &= \sup_{(x, y) \in I^2} \left| \int_0^x \int_0^y st (f(s, t)^2 - h(s, t)^2) ds dt \right| \\ &\leq \sup_{(x, y) \in I^2} \int_0^x \int_0^y st |f(s, t)^2 - h(s, t)^2| ds dt \\ &\leq \frac{1}{4} \|f - h\|_\infty \cdot \|f + h\|_\infty \\ &\leq \frac{1}{4} \cdot 2R \cdot \|f - h\|_\infty \end{aligned}$$

So we need to have $\frac{R}{2} < 1$ and we also need to have $g(f) \in A$ so

$$|g(f)| < |v| + \left| \int_0^x \int_0^y st f(s, t)^2 ds dt \right| < |v| + \frac{1}{4} R^2$$

Therefore, we also need $|v| + \frac{1}{4} R^2 < R$. Now let $R = 1$ and $|v| = 1/4$. Then by Banach Fixed point theorem, there exists $f \in A$ with $g(f) = f$, which is what we want. $\square$

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be 2π -periodic such that $|f(x+h) - f(x)| \leq |h|^\alpha$ for all $x \in \mathbb{R}$ and all $h \in \mathbb{R}$ and some $\alpha \in (0, 1)$. Show that

$$|\hat{f}| \leq C|n|^{-\alpha}$$

for some $C > 0$.

Proof.

$$\begin{aligned} \hat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} f(x) dx \\ &= \frac{1}{2\pi} \int_{-\pi-\pi/n}^{\pi-\pi/n} e^{-in(x+\pi/n)} f(x+\pi/n) dx \\ &= -\frac{1}{2\pi} \int_{-\pi-\pi/n}^{\pi-\pi/n} e^{-inx} f(x+\pi/n) dx \\ &= -\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} f(x+\pi/n) dx \quad \text{since } f \text{ is } 2\pi\text{-periodic} \end{aligned}$$

Therefore,

$$|\widehat{f}(n)| = \frac{1}{4\pi} \left| \int_{-\pi}^{\pi} e^{inx} (f(x) - f(x + \pi/n)) dx \right| \leq \frac{2\pi}{4\pi} \left(\frac{\pi}{n} \right)^{\alpha} = \frac{\pi^{\alpha}}{2} \cdot n^{-\alpha}$$

□

7. Skipped.

Week 8

1. Let $f(\theta) = |\theta|$ on $[-\pi, \pi]$. Compute the Fourier coefficients.

Proof.

$$\begin{aligned} \widehat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| e^{-inx} dx \\ &= \frac{1}{2\pi} \left(\int_0^{\pi} x e^{-inx} dx + \int_{-\pi}^0 (-x) e^{-inx} dx \right) \\ &= \frac{1}{2\pi} \int_0^{\pi} x (e^{inx} + e^{-inx}) dx \\ &= \frac{(-1)^n - 1}{\pi n^2} \quad n \neq 0 \end{aligned}$$

□

2. Question on Fejer Kernel.

3. Another question on Fejer Kernel.

4. Question on Sobolev-Embedding Theorem.

8. (Royden) Let B be the set of rational numbers in $[0, 1]$ and $\{I_k\}_{k=1}^n$ be a finite collection of open intervals that covers B . Prove that $\sum_{k=1}^n m^*(I_k) \geq 1$.

Proof. Suppose $\sum_{k=1}^n m^*(I_k) = c < 1$. Let $I_k = (a_k, b_k)$. If there exists $x \notin \bigcup_k \overline{I_k}$ then $x \in (0, 1) \cap \overline{I_k}^c$. Since the latter set is open, $\exists (a, b) \subset (0, 1) \cap \bigcap_k \overline{I_k}^c$ containing x . This interval contains a rational number q . Then $q \notin \bigcup_k \overline{I_k}$ is a contradiction. Hence, $\bigcup_k \overline{I_k}$ contains the interval $(0, 1)$. Therefore, $m^*(\bigcup_k \overline{I_k}) \geq m^*((0, 1))$. But

$$1 > c = \sum_k m^*(I_k) \geq m^*\left(\bigcup_k I_k\right) \geq m^*((0, 1)) = 1$$

is a contradiction hence $\sum_{k=1}^n m^*(I_k) \geq 1$.

□

10. (Royden) Let A, B bounded sets for which there exists an $\alpha > 0$ such that $|x - y| \geq \alpha$ for all $x \in A, y \in B$. Prove that $m^*(A \cup B) = m^*(A) + m^*(B)$.

Proof. Since $\{A, B\}$ is a cover of $A \cup B$, we already have

$$m^*(A \cup B) \leq m^*(A) + m^*(B)$$

So we need to prove

$$m^*(A \cup B) \geq m^*(A) + m^*(B)$$

Let $\varepsilon > 0$. There exists an open cover $\{I_k\}$ of $A \cup B$ such that

$$\sum l(I_k) < m^*(A \cup B) + \varepsilon$$

and that $l(I_k) < \alpha$ for every k . Then $I_k \cap A \cap B$ is empty for every k so we can group the open sets in the open cover into two groups.

$$S_1 = \{I_j \in \{I_k\}_k : I_j \cap A \neq \emptyset\}$$

$$S_2 = \{I_j \in \{I_k\}_k : I_j \cap B \neq \emptyset\}$$

$$S_3 = \{I_k\}_k \setminus (S_1 \cup S_2)$$

Then, by definition

$$m^*(A) + m^*(B) \leq \sum_{I_j \in S_1} l(S_1) + \sum_{I_j \in S_2} l(S_2) < \sum_k l(I_k) < m^*(A \cup B) + \varepsilon$$

Letting $\varepsilon \rightarrow 0$, we obtain the other inequality. $\square$

11. (Royden) Prove that if a σ -algebra contains intervals of the form (a, ∞) then it contains all intervals.

Proof.

$$\begin{aligned} (-\infty, b] &= (b, \infty)^c \\ [b, \infty) &= \bigcap_n (b - \frac{1}{n}, \infty) \\ (a, b) &= (a, \infty) \cap (-\infty, b) \end{aligned}$$

$\square$

Week 9

1. Let $B_{r,\varepsilon}$ be the set of Riemann integrable functions $f : [0, \varepsilon] \rightarrow \mathbb{R}$ such that $\|f\|_2 \leq r$. For any f , define the functions

$$P(f)(x) := \int_0^x f(t)^2 dt$$

Show that $\|P(f) - P(g)\|_\infty \leq 2(\|f\|_2 + \|g\|_2) \cdot \|f - g\|_2$ and so if $r, \varepsilon > 0$ are small enough then P is a contraction from $B_{r,\varepsilon}$ to $B_{r,\varepsilon}$.

Proof.

$$\begin{aligned}
\|P(f) - P(g)\|_\infty &= \sup_{x \in \mathbb{R}} \left| \int_0^x f(t)^2 - g(t)^2 dt \right| \\
&\leq \sup_{x \in \mathbb{R}} \int_0^x |f(t)^2 - g(t)^2| dt \\
&\leq \int_0^\varepsilon |f - g| \cdot |f + g| \\
&\leq \|f - g\|_2 \cdot \|f + g\|_2 \quad \text{Hölder's} \\
&\leq (\|f\|_2 + \|g\|_2) \cdot \|f - g\|_2
\end{aligned}$$

Now we can relate the 2-norm to the supremum norm.

$$\|f - g\|_2 = \left(\int_0^\varepsilon |f - g|^2 \right)^{1/2} \leq \sqrt{\varepsilon} \cdot \|f - g\|_\infty$$

So for $2r\sqrt{\varepsilon} < 1$, we see that

$$\|P(f) - P(g)\|_\infty < 2r \cdot \sqrt{\varepsilon} \cdot \|f - g\|_\infty$$

so P is a contraction. □

2. Let $f : [-\pi, \pi] \rightarrow \mathbb{R}$ be a Riemann integrable function. Set

$$\widehat{f}(n) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

Define $u(x, t) : [-\pi, \pi] \times (0, \infty) \rightarrow \mathbb{C}$ by

$$u(x, t) = \sum_{n=0}^{\infty} \widehat{f}(n) e^{-n^2 t} e^{inx}$$

Show that the series above is absolutely convergent and that $u(x, t)$ is smooth and solves the heat equation $u_t = u_{xx}$. Finally, show that $\lim_{t \rightarrow 0^+} \|u(t, \cdot) - f\|_2 = 0$.

Proof. For absolute convergence, notice that

$$\begin{aligned}
\sum_{n=0}^{\infty} |\widehat{f}(n) e^{-n^2 t} e^{inx}| &\leq \sum_{n=0}^{\infty} \frac{e^{-n^2 t}}{2\pi} \int_{-\pi}^{\pi} |f(x)| dx \\
&\leq M \sum_{n=0}^{\infty} e^{-n^2 t} \\
&< \infty
\end{aligned}$$

The last inequality is true because $\int_0^\infty e^{-x} dx < \infty$. Since the series converge absolutely, we can differentiate term by term to get

$$\begin{aligned}
\partial_t u &= -n^2 u(x, t) \\
\partial_x^2 u &= -n^2 u(x, t)
\end{aligned}$$

so u solves the heat equation. Finally,

$$u(x, t) - f(x) = \sum_{n \in \mathbb{Z}} (e^{-n^2 t} - 1) \hat{f}(n) e^{-inx}$$

so the Fourier coefficients of $u(x, t) - f(x)$ is $(e^{-n^2 t} - 1) \hat{f}(n)$. Then by Parseval's identity,

$$\|u(\cdot, t) - f\|_2^2 \leq 2\pi \sum_{n \in \mathbb{Z}} |(e^{-n^2 t} - 1) \hat{f}(n)|^2 \xrightarrow{t \rightarrow 0} 0$$

The latter series converges by the M test given above. □

3. Let (X, d) be a compact metric space. Show that the functions of the form

$$F(x, y) = \sum_{j=1}^K f_j(y) g_j(x)$$

for some $K \in \mathbb{Z}$ and $f_j, g_j : X \rightarrow \mathbb{R}$ continuous are dense in the space of continuous functions from $X \times X \rightarrow \mathbb{R}$.

Proof. We invoke Stone-Weierstrass theorem so we need to show this family separates points and vanishes nowhere. Let $(x, y) \in X \times X$, then take $f_1(x) = g_1(x) = 1$. Then $f_1(x)g_1(x)$ is non-vanishing so this family of functions vanishes nowhere. Given $(a, b), (p, q) \in X \times X$, with $a \neq p$, consider the function $f(x) = x - a$, $g(x) = 1$. Then $f(a)g(b) = 0$ and $f(p)g(q) = p - a \neq 0$ so this family of functions separates points as well. It is an algebra since such functions are closed under scalar multiplication, addition and multiplication. By Stone-Weierstrass, such family of functions are dense in the space of continuous functions from $X \times X \rightarrow \mathbb{R}$. □

4. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous. Show that there exists a sequence of polynomials in the variables x, y converging point-wise to F .

Proof. In the previous problem, let $X_n = [-n, n]$ and f_j, g_j can be approximated uniformly by polynomials in X_n , hence there exists a polynomial $p_n(x, y)$ in x and y variables such that

$$\|F - p_n\|_\infty < \frac{1}{n}$$

on X_n . Then the sequence $(p_n)_n$ converges to F pointwise. Let $(x, y) \in \mathbb{R}^2$ and $\varepsilon > 0$ be given. Then there exists $N \in \mathbb{N}$ such that $(x, y) \in X_N$ and $\frac{1}{N} < \varepsilon$ and for all $n > N$, $(x, y) \in X_n$ and

$$\|F - p_n\|_\infty < \frac{1}{n} < \frac{1}{N} < \varepsilon \implies |F(x, y) - p_n(x, y)| < \varepsilon$$

so $\lim_{n \rightarrow \infty} p_n(x, y) = F(x, y)$. □

5. Let V be a normed linear space and $F : V \rightarrow V$ a contraction. Show that $G(x) := x - F(x)$ is a bijection with a continuous inverse.

Proof. Let $v \in V$, define $h_v : V \rightarrow V$ by $h_v(x) = F(x + v)$. Then $h_v(x - v) = F(x)$ and $g(x) = v \iff x - v = F(x) \iff h_v(x - v) = x - v$. Notice that since F is a contraction,

$$|h_v(x) - h_v(y)| = |F(x + v) - F(y + v)| < \alpha \cdot |x - y|$$

By Banach fixed point theorem, h has a fixed point. Say $h_v(w) = w$ then $F(w + v) = w \implies v = (w + v) - F(w + v) = g(w + v)$ so g is surjective. $g(x)$ is also injective since the fixed point of F is unique. Notice that

$$\begin{aligned} |(v - F(v)) - (w - F(w))| &= |(v - w) - (F(v) - F(w))| \\ &\geq |v - w| - |F(v) - F(w)| \\ &> (1 - \alpha)|v - w| \end{aligned}$$

So

$$|x - y| \geq (1 - \alpha) \cdot |g^{-1}(x) - g^{-1}(y)|$$

which proves g^{-1} is continuous around x . □

6. Let $f_n : [-1, 1] \rightarrow \mathbb{R}$ be a twice differentiable function such that $f_n(0) = f'_n(0) = 0$ and $|f''_n(x)| \leq 1$ for all $x \in (-1, 1)$. Show that there exists a subsequence that converges uniformly to a differentiable function.

Proof.

$$|f_n(x)| \leq |f_n(0)| + \int_0^x |f'_n(t)| dt$$

On the other hand,

$$\begin{aligned} |f'_n(t)| &< |f'_n(0)| + \int_0^t |f''_n(s)| ds \\ &\leq 1 + |t| \\ &\leq 2 \end{aligned}$$

Therefore,

$$|f_n(x)| \leq 2|x| < 2$$

so f_n are uniformly bounded. Notice that

$$|f_n(x) - f_n(y)| = |f'_n(t)| \cdot |x - y| \leq 2|x - y|$$

for some $t \in (x, y)$ so f_n are equicontinuous as well. Then by Arzela-Ascoli, there exists a convergent subsequence. □

7. If $f : [-\pi, \pi] \rightarrow \mathbb{R}$ is real and Riemann integrable, show that its Fourier series is a real function.

Proof. Since $f(x) = \sum_{n \in \mathbb{Z}} c_n e^{inx}$. It suffices to prove that $\overline{c_n} = c_{-n}$.

$$\begin{aligned}\overline{c_n} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \overline{f(x)} e^{inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx \\ &= c_{-n}\end{aligned}$$

□

Week 10

1. Let $E \in \mathcal{E}$ satisfy $m(E) > 0$. Show that there is a bounded set $A \subseteq E$ such that $m(A) > 0$.

Proof. We can write

$$E = \bigcup_{n=1}^{\infty} (E \cap [-n, n])$$

Notice that $E \cap [-n, n]$ is an increasing sequence of nested sets hence

$$m^*(E) = \lim_{n \rightarrow \infty} m^*(E \cap [-n, n]) > 0$$

so for some n , $m^*(E \cap [-n, n]) > \frac{m^*(E)}{2} > 0$ and we can take $A = E \cap [-n, n]$. □

2. Show that every open set is a countable union of disjoint intervals.

Proof. Let O be a non-empty open set in $\mathbb{R}$ and let $x \in O$. Set

$$\begin{aligned}a_x &= \inf\{z : (z, x) \subset O\} \\ b_x &= \sup\{y : (x, y) \subset O\}\end{aligned}$$

$(a_x, b_x) \subset O$ and $a_x, b_x \notin O$ by openness of O . Then, if $y \notin (a_x, b_x)$, $(a_y, b_y) \cap (a_x, b_x) = \emptyset$. Otherwise, $a_x = a_y, b_x = b_y$ by construction. Then, letting I_x denote (a_x, b_x) , we have

$$O = \bigcup_{x \in O} I_x$$

with I_x occurring in the union once. Since each I_x contains a rational number, this union is countable. □

3. Let $E \in \mathcal{E}$, $m(E) > 0, 0 < \alpha < 1$. Show that there exists a bounded interval I so that

$$m(E \cap I) \geq \alpha \cdot m(I)$$

Proof. Suppose that for any bounded interval I ,

$$m(E \cap I) < \alpha \cdot m(I)$$

Choose $r \in (1, \frac{1}{\alpha})$. Then by the definition of measure, there exists a disjoint open cover $\{I_k\}$ of E such that

$$r \cdot m^*(E) > \sum_k l(I_k)$$

But then

$$\begin{aligned} r \cdot m^*(E) &> \sum_k l(I_k) \\ &> \sum_k \frac{1}{\alpha} \cdot m^*(I_k \cap E) \\ &\geq \frac{1}{\alpha} m^*\left(\bigcup (I_k \cap E)\right) \\ &= \frac{1}{\alpha} \cdot m^*(E) \end{aligned}$$

which implies $r > \frac{1}{\alpha}$, a contradiction. $\square$

4. Let $E \in \mathcal{E}$ have positive measure. Prove that the set $A := \{x - y : x, y \in E\}$ contains an open interval J such that $0 \in J$.

Proof. Suppose $t \in E - E$. This means $E \cap (E + t) \neq \emptyset$. So we want to find an interval J containing 0 such that $E \cap (E + t) \neq \emptyset$ for every $t \in J$. By the previous problem, there exists a bounded interval I such that

$$m(E \cap I) > \frac{3}{4}m(I)$$

Set $J = (-\frac{m(I)}{2}, \frac{m(I)}{2})$. Suppose there exists some $t \in J$ such that $E \cap (E + t) = \emptyset$. Then $I \cap (E \cap (E + t)) = (E \cap I) \cap (E \cap I + t) = \emptyset$ as well. However,

$$\frac{3}{4}m(I) < m(E \cap I) = \frac{1}{2}m(E \cap I) + \frac{1}{2}m(E \cap I + t) \leq \frac{1}{2}m(I) + \frac{1}{4}m(I) = \frac{3}{4}m(I)$$

is a contradiction. Hence $J \subset E - E$. $\square$

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Lebesgue measurable function such that

$$f(x + y) = f(x) + f(y)$$

Show that $f(x) = ax$ for some $a \in \mathbb{R}$.

Proof. Firstly, notice that $f(nx) = nf(x)$ for $n \in \mathbb{N}$ so $\frac{1}{n}f(x) = f(\frac{x}{n})$ and $f(\frac{a}{b}x) = af(\frac{x}{b}) = \frac{a}{b}f(x)$. Hence $f(q) = qf(1)$ for every $q \in \mathbb{Q}$. Secondly, observe that $f(0) = 0$. The idea is to show that f is continuous around 0. If that is the case, then $\lim_{h \rightarrow 0} f(x + h) = \lim_{x \rightarrow 0} f(x) + f(h) = f(x)$ implies that f is continuous everywhere. Then $f(x) = xf(1)$ for every $x \in \mathbb{R}$ by the denseness of the rationals. We apply Lusin's theorem and the previous question to prove continuity at 0. By Lusin's theorem, there exists a compact set $K \subset [-1, 1]$ such that

$$m^*([-1, 1] \setminus K) < \frac{1}{5}$$

Then K is measurable since it is closed. Moreover,

$$m^*(K) = m([-1, 1]) - m^*([-1, 1] \setminus K) > 0$$

The above equality is from the measurability of K . K might not contain 0 yet but we can apply the previous problem to find an interval $(-\delta, \delta) \subset K - K$. Then for $h \in K - K$ we can write $h = x - y$, where $x, y \in K$, and

$$\lim_{h \rightarrow 0} f(h) = \lim_{x \rightarrow y} f(x - y) = \lim_{x \rightarrow y} f(x) - f(y) = 0$$

This shows f is continuous at 0 and effectively proves the claim. $\square$

19. (Royden) Let E have finite outer measure. Show that if E is not measurable, then there is an open set O containing E that has finite outer measure and for which

$$m^*(O \setminus E) > m^*(O) - m^*(E)$$

Approximation by open sets. Suppose such a set does not exist. Then for any open set O containing E with finite outer measure ,

$$m^*(O \setminus E) \leq m^*(O) - m^*(E)$$

Moreover, since $\{O \setminus E, E\}$ covers O , we have

$$m^*(O) \leq m^*(E) + m^*(O \setminus E) \implies m^*(O \setminus E) \geq m^*(O) - m^*(E)$$

This shows that for any such O ,

$$m^*(O \setminus E) = m^*(O) - m^*(E)$$

Given $\varepsilon > 0$, there exists an open disjoint cover $\{I_k\}$ of E such that

$$\sum_k l(I_k) < m^*(E) + \varepsilon$$

Then taking $O = \bigcup I_k$, we have

$$m^*(O \setminus E) = m^*(O) - m^*(E) \leq \sum_k l(I_k) - m^*(E) < \varepsilon$$

The characterization of measurable sets via outer approximation by open sets tells us that E is measurable, a contradiction. $\square$

20. (Royden) Let E have finite outer measure. Show that E is measurable if and only if for each open bounded interval (a, b) ,

$$b - a = m^*((a, b) \cap E) + m^*((a, b) \setminus E)$$

Proof. Let A be a set. We want to prove

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E)$$

If $m^*(A) = \infty$, it is automatically true so let $m^*(A) < \infty$. Then, $m^*(A) \leq m^*(A \cap E) + m^*(A \setminus E)$ is true since $\{A \cap E, A \setminus E\}$ covers A . For the other inequality, we prove

$$m^*(A) + \varepsilon \geq m^*(A \cap E) + m^*(A \setminus E)$$

for every $\varepsilon > 0$. Given $\varepsilon > 0$, there exists a disjoint open cover $\{I_k\}$ of A such that

$$l(I_k) < m^*(A) + \varepsilon$$

Then

$$\begin{aligned} m^*(A) + \varepsilon &> \sum_k l(I_k) \\ &= \sum_k m^*(I_k \cap E) + m^*(I_k \setminus E) \\ &\geq m^*\left(\bigcup_k I_k \cap E\right) + m^*\left(\bigcup_k I_k \setminus E\right) \\ &\geq m^*(A \cap E) + m^*(A \setminus E) \end{aligned}$$

which finishes the proof. $\square$

4. (Royden) Suppose f is a real valued function on $\mathbb{R}$ such that $f^{-1}(c)$ is measurable for each number c . Is f necessarily measurable?

Proof. No. Let E be a non-measurable set. Consider

$$f(x) = \begin{cases} e^x & x \in E \\ -e^x & x \notin E \end{cases}$$

This function is one-to-one so every preimage contains at most one point, hence is measurable. However, $f^{-1}((0, \infty)) = E$ is not a measurable set, hence f is not measurable. $\square$

6. (Royden) Let $f : D \rightarrow \mathbb{R}$ be a function where D is measurable. Show that f is measurable if and only if the function g defined on $\mathbb{R}$ by $g(x) = f(x)$ if $x \in D$ and $g(x) = 0$ for $x \notin D$ is measurable.

Proof. If g is measurable then f is obviously measurable. Conversely, let f be measurable. Consider the open set (c, ∞) . If $c > 0$ then

$$g^{-1}((c, \infty)) = f^{-1}((c, \infty))$$

is measurable. If $c < 0$, then

$$g^{-1}((c, \infty)) = g^{-1}((c, 0)) \cup g^{-1}(0) \cup g^{-1}((0, \infty)) = f^{-1}((c, 0)) \cup D^c \cup f^{-1}((0, \infty))$$

all of which are measurable since D is measurable, and f is measurable. Thus, $g^{-1}((c, \infty))$ is measurable in this case too. Hence g is measurable. $\square$

13. (Royden) A real-valued measurable function is said to be semisimple provided it only takes a countable number of values. Let f be any measurable function on E . Show that there exists a sequence of semisimple functions $\{f_n\}$ on E that converges uniformly to f on E .

Proof. The idea is to construct functions f_n satisfying

$$|f(x) - f_n(x)| < \frac{1}{n} \quad \forall x \in E$$

To construct f_n , partition $(-\infty, \infty)$ into intervals of length $\frac{1}{n}$

$$\dots, [-\frac{1}{n}, 0), [0, \frac{1}{n}), [\frac{1}{n}, \frac{2}{n}), \dots$$

and set $I_k = [\frac{k-1}{n}, \frac{k}{n})$ and $J_k = [-\frac{k}{n}, -\frac{(k-1)}{n})$. Since f is measurable, $E_k = f^{-1}(I_k)$, $F_k = f^{-1}(J_k)$ are measurable sets. Now set

$$f_n = \sum_{k=1}^{\infty} (-k) \cdot \chi_{F_k} + (k-1) \cdot \chi_{E_k}$$

Given $x \in E$, $f(x) \in I_k$ or $f(x) \in J_k$ for some k . In the first case $f_n(x) = -k$, in the second case $f_n(x) = k-1$, either way $|f_n(x) - f(x)| < \frac{1}{n}$. Therefore $f_n \rightarrow f$ uniformly on E . $\square$

Week 11

27. (Royden) Show that Egoroff's theorems is no longer true if the drop the assumption that E has finite measure.

Proof. Consider the hat function f_n where f_n on $[0, 1]$ divides the interval into n equal pieces and on each piece f_n looks like a hat with a peak $\frac{1}{n}$. Extend f_n to the whole real line by multiplying the function on $[0, 1]$ by a factor of $|m|$ on the intervals $[m, m+1)$ and $[-m, -m+1)$. Then $f_n \rightarrow 0$ pointwise on $\mathbb{R}$ but not uniformly. If F is a closed set with $m^*(\mathbb{R} \setminus F) < \varepsilon$, then $m^*(F) = \infty$ so F is not bounded. On unbounded subsets of $\mathbb{R}$, the functions f_n do not converge to 0 uniformly. The Egoroff theorem does not hold. Indicator function $f_n = \chi_{[-n, n]}$. $\square$

29. (Royden) Prove Lusin's theorem when $m^*(E) = \infty$.

Proof. Let $\varepsilon > 0$. Apply finite Lusin to $E \cap [-n, n]$ to get a continuous function $g_n : \mathbb{R} \rightarrow \mathbb{R}$ that agrees with f on a closed set $F_n \subset E \cap [-n, n]$ where

$$m^*(E \setminus F) < \frac{\varepsilon}{2^{n+1}}$$

Then consider $F = \bigcup_n F_n$.

$$m^*(E \setminus F) = m^*(E \setminus \bigcup_n F_n) = m^*(\bigcup_n (E \setminus F_n)) \leq \sum_{n=1}^{\infty} m^*(E \setminus F_n) < \varepsilon/2$$

To construct a consistent function g , let $\tilde{F}_n := F_n \setminus \bigcup_{k < n} F_k$. Then $\bigcup_n \tilde{F}_n = F$ and $\tilde{F}_n$ are disjoint. Now define $g(x) := g_n(x)$ where $x \in \tilde{F}_n$. Then $f = g$ on F since given $x \in F$, $x \in \tilde{F}_n \subset F_n$ for some n and $g(x) = g_n(x) = f(x)$. Finally, since F is measurable, we can find a closed set $F_0 \subset F$ such that

$$m^*(F \setminus F_0) < \varepsilon/2$$

Then we have

$$m^*(E \setminus F_0) \leq m^*(E \setminus F) + m^*(F \setminus F_0) < \varepsilon$$

as desired. $\square$

20. (Royden) Let $\{f_n\}$ be a sequence of non-negative measurable functions that converges to f point-wise on E . Let M be such that $\int_E f_n \leq M$ for all n . Show that $\int_E f \leq M$. Verify that this property is equivalent to the statement of Fatou's lemma.

Proof. Let h be a bounded measurable function of finite support with $0 \leq h \leq f$. Let $E_0 = \{x \in E : h(x) \neq 0\}$. Define $h_n = \min(f_n, h)$ on E . Then $h_n \equiv 0$ on $E \setminus E_0$. Since $\{f_n\} \rightarrow f$ pointwise and $h_n \leq h$, we have that $\{h_n\} \rightarrow h$ pointwise. Moreover, since h is bounded, say $|h| \leq N$ for some N . Then $|h_n| \leq N$ as well. So h_n are uniformly bounded. Since h has finite support $m^*(E_0) < \infty$. Then by the Bounded Convergence theorem applied to the functions $\{h_n\}$

$$\lim_{n \rightarrow \infty} \int_E h_n = \int_E h$$

and

$$h_n \leq f_n \implies \int_E h_n \leq \int_E f_n \leq M \implies \int_E h \leq M$$

Since this is true for every such function h , the supremum over $\int_E h$ is $\int_E f$. Therefore,

$$\int_E f \leq M$$

$\square$

21. (Royden) Let the function f be non-negative and integrable over E and $\varepsilon > 0$. Show that there is a simple function η on E that has finite support, $0 \leq \eta \leq f$ on E and $\int_E |f - \eta| < \varepsilon$. If E is a closed bounded interval, show that there is a step function h on E that has finite support and $\int_E |f - h| < \varepsilon$.

Proof. Since

$$\int_E f := \sup \left\{ \int_E h : 0 \leq h \leq f, h \text{ bounded, measurable of finite support} \right\}$$

there exists such h such that

$$\int_E f - \varepsilon/2 < \int_E h$$

Say h is supported on E_0 . Then $m^*(E_0) < \infty$ so by definition of the integral, there exists a simple function φ on E_0 such that $0 \leq \varphi \leq h$ and

$$\int_E h - \varepsilon/2 < \int_{E_0} \varphi$$

Then

$$\int_E (f - \varphi) = \int_E (f - h + h - \varphi) < \varepsilon$$

If E is closed and bounded, then it is contained in an interval of the form $[a, b]$. Over such intervals, Riemann integral is the same as the Lebesgue integral hence $\int_E f$ can be approximated via step functions. $\square$

25. (Royden) Let f_n be a sequence of non-negative measurable functions on E that converges point-wise on E to f . Suppose $f_n \leq f$ on E for each n . Show that

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$$

Proof. Since $f_n \leq f$, $\int_E f_n \leq \int_E f$ so

$$\lim_{n \rightarrow \infty} \int_E f_n \leq \int_E f$$

To prove the other inequality, let h be a bounded measurable function of finite support such that $0 \leq h \leq f$ and set $h_n := \min(f_n, h)$. Since h is bounded, h_n are uniformly bounded and since $f_n \rightarrow f$ pointwise, $h_n \rightarrow h$ pointwise. Then by the bounded convergence theorem,

$$\int_E h = \lim_{n \rightarrow \infty} \int_E h_n \leq \lim_{n \rightarrow \infty} \int_E f_n$$

The supremum over the expressions on the left hand side is precisely $\int_E f$ so

$$\int_E f \leq \lim_{n \rightarrow \infty} \int_E f_n$$

as desired. □

1. Let $E \in \mathcal{E}$ have positive measure. Prove that there exists $p \in E$ such that

$$m^*(E \cap \{x : |x - p| < \varepsilon\}) > 0$$

for every $\varepsilon > 0$.

Proof. Suppose not. Then, for every $p \in E$ there exists ε_p such that

$$m^*(E \cap \{x : |x - p| < \varepsilon_p\}) = 0$$

Then $I_p = (p - \varepsilon_p, p + \varepsilon_p)$ cover E . Pass to a countable subcover $\{I_n\}_{n=1}^\infty$ of $\{I_p\}_{p \in E}$ then

$$0 < m^*(E) \leq m^*\left(\bigcup_{k=1}^\infty (I_k \cap E)\right) \leq \sum_{k=1}^\infty m^*(I_k \cap E) = 0$$

is a contradiction. □

2. Let $f : E \rightarrow [0, \infty]$ satisfy $\int_E f < \infty$. Let $A_n \subset A_{n-1} \subset \cdots \subset E$ be a decreasing sequence of sets such that $\lim_{n \rightarrow \infty} m(A_n) = 0$. Prove that

$$\lim_{n \rightarrow \infty} \int_E f \cdot \chi_{A_n} = 0$$

Lebesgue Dominated Convergence Theorem. Let $f_n = f \cdot \chi_{A_n}$. Then, notice that $|f_n| \leq f$ for every n and since f is integrable, we can apply the Dominated Convergence Theorem. Since $\{A_n\}$ is a decreasing sequence of sets, $f_n \rightarrow 0$ point-wise. Then it follows that

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E 0 = 0$$

which proves the claim. □

3. Let $f_n : E \rightarrow [0, \infty]$ be a decreasing sequence of Lebesgue measurable functions such that $\int_E f_1 < \infty$. Let f be the point-wise limit. Show that

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$$

Proof. The sequence $\{f_1 - f_n\}$ converges pointwise to $f_1 - f$ and it is an increasing sequence since

$$f_1 - f_{n+1} \geq f_1 - f_n \iff f_n \geq f_{n+1}$$

Moreover, $f_1 - f_n$ is non-negative for any n . By Monotone Convergence theorem,

$$\lim_{n \rightarrow \infty} \int_E (f_1 - f_n) = \int_E f_1 - f \implies \lim_{n \rightarrow \infty} \int_E f_n = \int_E f$$

as desired. □

4. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is integrable, show that the function

$$F(x) = \int_{\mathbb{R}} 1_{(-\infty, x)} f$$

is continuous.

Proof. Let $x \in \mathbb{R}$ and $\varepsilon > 0$ be given. Set

$$A_n := \{y \in \mathbb{R} : f(y) \geq n\}$$

Notice that $\{A_n\}$ is a decreasing sequence of sets and since $|f|$ is integrable, $\lim_{n \rightarrow \infty} m^*(A_n) = 0$. Then by question 2, there exists N such that for all $n \geq N$,

$$\int_{\mathbb{R}} |f| \cdot 1_{A_n} < \varepsilon/2$$

Choose $\delta = \frac{\varepsilon}{2N}$. Then, given $|x - y| \leq \delta$, wlog assume $y \leq x$, we have

$$\begin{aligned} |F(x) - F(y)| &\leq \int_y^x |f| \\ &= \int_{[y, x] \cap A_N} |f| + \int_{[y, x] \cap A_N^c} |f| \\ &\leq \int_{[y, x] \cap A_N} |f| + |x - y| \cdot N \\ &\leq \frac{\varepsilon}{2} + \delta N \\ &= \varepsilon \end{aligned}$$

□

Week 12

28. (Royden) Prove that if f is integrable over E and $C \subset E$ is a measurable set

$$\int_C f = \int_E f \cdot \chi_C$$

Proof. It suffices to prove that

$$\int_E f^+ \cdot \chi_C = \int_C f^+ \cdot \chi_C$$

since $(f \cdot \chi_C)^+ = f^+ \cdot \chi_C$. Let h be a measurable bounded function finitely supported in C , and $0 \leq h \leq f^+$. Then extend h to E by setting $h(x) = 0$ if $x \notin C$. It follows that

$$h \leq f^+ \cdot \chi_C$$

so

$$\int_C h = \int_E h \leq \int_E f^+ \cdot \chi_C$$

The supremum over the integrals on the left hand side is precisely $\int_C f$ so we get

$$\int_C f^+ \leq \int_E f^+ \cdot \chi_C$$

Conversely, let $0 \leq h \leq f^+ \cdot \chi_C$ be a measurable bounded function finitely supported on E . The inequality relations imply that $h = 0$ outside C so h restricts to a measurable bounded function finitely supported on C . This gives the other way inequality. $\square$

29. (Royden) For a measurable function f on $[1, \infty)$ define $a_n = \int_n^{n+1} f$. Is it true that f is integrable if and only if the sum $\sum_{n=1}^{\infty} a_n$ converges? What if it absolutely converges?

Proof. It is not enough for the series $\sum_{n=1}^{\infty} a_n$ to converge. The integral is defined as

$$\int_1^{\infty} f = \int_1^{\infty} f^+ - \int_1^{\infty} f^-$$

so if $a_n = (-1)^n \frac{1}{n}$ then $\sum_{n=1}^{\infty} a_n$ converges but the integral is given by $\sum_{n=2k}^{\infty} a_n + \sum_{n=2k+1}^{\infty} a_n$ which does not converge. If $\sum_n a_n$ is absolutely convergent then rearranging does not change the value of the integral so f is integrable then. $\square$

32. (Royden) **General Lebesgue Dominated Convergence Theorem** Let $\{f_n\}$ be a sequence of measurable functions on E that converges pointwise almost everywhere on E to f . Suppose there is a sequence $\{g_n\}$ of non-negative measurable functions on E that converges pointwise almost everywhere on E to g and dominates $\{f_n\}$ in the sense that

$$|f_n| \leq g_n \text{ on } E \text{ for all } n.$$

If $\lim_{n \rightarrow \infty} \int_E g_n = \int_E g$, then

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$$

Proof. Since $|f_n| \leq g_n$, it is measurable. Then by possibly excising a countable union of sets of measure 0 from E , we can assume that f and f_n are finite on E . Hence the functions $g_n - f_n$ and $g_n - f$ are properly defined and they converge point-wise to $g - f$ almost everywhere on E . By Fataou's lemma,

$$\int_E (g - f) \leq \liminf \int_E g_n - f_n$$

By linearity of the integral,

$$\begin{aligned} \int_E (g - f) &= \int_E g - \int_E f \\ &\leq \liminf \int_E g_n - \limsup \int_E f_n \\ &= \lim \int_E g_n - \limsup \int_E f_n \\ &= \int_E g - \limsup \int_E f_n \end{aligned}$$

which implies

$$\limsup \int_E f_n \leq \int_E f$$

By considering $\{g_n + f_n\}$ and applying the same procedure, we see that

$$\int_E f \leq \liminf \int_E f_n$$

□

2. Let f_n be a sequence of non-negative functions such that $f_n \geq f_{n+1}$ and $\int_{\mathbb{R}} f_1 < \infty$. Prove that

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E \lim_{n \rightarrow \infty} f_n$$

Proof. Let $x \in \mathbb{R}$. The set $\{f_n(x) : n \in \mathbb{N}\}$ is bounded below by 0 and is a decreasing sequence hence it converges to its infimum. Then $f(x) = \inf_n f_n(x)$ is the point-wise limit function. $\{f_n\}$ is dominated by f_1 , which is integrable, so by the Lebesgue Dominated Convergence theorem,

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E \lim_{n \rightarrow \infty} f_n$$

□

4. For $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ define

$$\|f\|_{\infty} := \inf \{s : m(\{x : |f(x)| > s\}) = 0\}$$

where $\inf \emptyset := \infty$. Let $L^{\infty}(\mathbb{R})$ be the set of equivalence classes of f where $\|f\|_{\infty} < \infty$. Prove that $|f(x)| \leq \|f\|_{\infty}$ for almost all x on E and that L^{∞} is complete.

Proof. Let $A := \{s : m^*(x : |f(x)| > s) = 0\}$. By definition of $\|f\|_\infty$, for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that

$$a_n < \|f\|_\infty + \frac{1}{n}$$

Since $a_n \in A$, if we define $E_n := \{x : |f(x)| > a_n\}$, we have $m^*(E_n) = 0$. And $|f(x)| \leq a_n$ on $E \setminus E_n$. Hence

$$|f| < \|f\|_\infty + \frac{1}{n}$$

on $E \setminus E_n$. Let $E_0 = \bigcup_{n=1}^\infty E_n$. Then $m^*(E_0) = 0$ and

$$|f(x)| < \|f\|_\infty$$

on $E - E_0$. Let $\{f_n\}$ be a Cauchy sequence in $L^\infty(\mathbb{R})$. Then notice that for fixed $x \in E$,

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty \xrightarrow{n, m \rightarrow \infty} 0$$

for almost every x . Let this inequality be false on the set E_0 . Then on $E \setminus E_0$ the above inequality holds so $\{f_n(x)\}_{x \in E \setminus E_0}$ forms a Cauchy Sequence in $\mathbb{R}$. Since $\mathbb{R}$ is complete, it converges to a number x^* . Define f on $E \setminus E_0$ this way and define f to be 0 on E_0 . Then it is obvious that

$$|f_n(x) - f(x)| < \varepsilon$$

for all large enough n for all $x \notin E_0$. Then this shows that

$$\|f - f_n\|_\infty < \varepsilon$$

for large enough n as well. Hence

$$\lim_{n \rightarrow \infty} \|f - f_n\|_\infty = 0$$

□

5. Let E be a Lebesgue measurable set with $m(E) > 0$. Let $f_n : E \rightarrow \mathbb{R}$ be a sequence of Lebesgue measurable functions. We say that $f_n \rightarrow f$ in measure if for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} m(\{x : |f(x) - f_n(x)| > \varepsilon\}) = 0.$$

- (a) If a sequence converges in measure show that a subsequence converges a.e.
- (b) If $m(E) < \infty$ show that a pointwise convergent sequence converges in measure.
- (c) Find an example of a sequence that converges in measure but not pointwise.
- (d) If $f_n \rightarrow f$ in L^p show that $f_n \rightarrow f$ in measure.

Proof. For $\varepsilon = \frac{1}{k}$ choose n_k such that

$$m(\{x : |f_{n_k}(x) - f(x)| > \frac{1}{k}\}) < \frac{1}{2^k}$$

and let $E_{m,n} := \{x : |f_m(x) - f(x)| > \frac{1}{n}\}$. With this definition, we have that

$$m(E_{n_k, k}) \leq \frac{1}{2^k}$$

$E_{n_k, k}$ is a bad set on which f_{n_k} is not quite close enough to f as we like. But $\sum_k 2^{-k} < \infty$ so the total badness is summable. Let $F_\ell = \bigcup_{k \geq \ell} E_{n_k, k}$. For $k > \ell$ observe that

$$E_{n_k, k} \subset E_{n_k, l} \subset E_{n_l, l}$$

so $m(F_\ell) = \lim_{k \rightarrow \infty} m(E_{n_k, k}) \leq \lim_{k \rightarrow \infty} \frac{1}{2^k} = 0$. Define $F = \bigcap_{\ell=1}^{\infty} F_\ell$. Then $m^*(F) = 0$ as well. If $x \notin F$, observe that there exists ℓ such that $x \notin F_\ell$ hence $x \notin E_{n_k, k}$ for every $k > \ell$. Hence,

$$|f_{n_k}(x) - f(x)| < \frac{1}{k}$$

for every $k > \ell$. Therefore, $f_{n_k}(x) \rightarrow f(x)$ as $k \rightarrow \infty$ for every $x \in E$. This proves part a.

For part b, assume $f_n \rightarrow f$ pointwise and suppose that there exists $\varepsilon > 0$ such that

$$\lim_{n \rightarrow \infty} m(\{x : |f(x) - f_n(x)| > \varepsilon\}) > 0$$

Then define $E_n = \{x : |f(x) - f_n(x)| > \varepsilon\}$. Since $f_n \rightarrow f$ pointwise, for every x , there exists a minimal $m_x \in \mathbb{N}$ such that

$$|f_m(x) - f(x)| < \varepsilon \quad \forall m \geq m_x$$

Observe that $E_n = \{x : m_x > n\}$. This characterization of E_n makes it easy for us to see that $E_{n+1} \subset E_n$. Now set $E = \bigcap_{n=1}^{\infty} E_n$. Then it follows that $m(E) = \lim_{n \rightarrow \infty} m(E_n) > 0$ by assumption. For $x \in E$, $f_n(x) \not\rightarrow f(x)$ since $x \in E_n$ for all n so

$$|f_n(x) - f(x)| > \varepsilon$$

for every n . But then f_n does not converge to f almost everywhere.

Take the typewriter sequence for part c. It converges in measure but not pointwise.

Suppose $f_n \rightarrow f$ in L^p . Then by Chebyshev's inequality,

$$\begin{aligned} m(\{x : |f_n(x) - f(x)| > \varepsilon\}) &= m(\{x : |f_n(x) - f(x)|^p > \varepsilon^p\}) \\ &< \frac{1}{\varepsilon^p} \int |f_n - f|^p \\ &\xrightarrow{n \rightarrow \infty} 0 \end{aligned}$$

Hence $f_n \rightarrow f$ in measure as well. □

6. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function.

(a) Let P_k be a sequence of partitions of $[a, b]$ such that P_{k+1} is a refinement of P_k , and the mesh size goes to 0. For the partitions P_k define the functions

$$G_k(x) = \sum_j M_j \mathbf{1}_{[t_{j-1}, t_j)}(x), \quad g_k(x) = \sum_j m_j \mathbf{1}_{[t_{j-1}, t_j)}(x),$$

with

$$M_j = \sup_{x \in [t_{j-1}, t_j]} f(x), \quad m_j = \inf_{x \in [t_{j-1}, t_j]} f(x).$$

Show that the Lebesgue integrals of $G_k(x)$ and $g_k(x)$ are the upper and lower Riemann sums of f associated to the partition P_k .

(b) Show that the limits $G(x) = \lim_{k \rightarrow \infty} G_k(x)$ and $g(x) = \lim_{k \rightarrow \infty} g_k(x)$ exist and that

$$g \leq f \leq G.$$

(c) If f is Riemann integrable, show that $g = G$ a.e.

(d) If f is Riemann integrable, deduce that it is Lebesgue measurable and that its Riemann integral equals its Lebesgue integral.

Proof. This question is easy. □

Final Practice

1. Let $m(E) > 0$ and $f_n \rightarrow f$ in measure. Prove that if $f_n \geq 0$ then $\liminf \int_E f_n \geq \int_E f$. Suppose now that $|f_n| \leq g$ for some integrable g , prove the same statement.

Proof. If $f_n \rightarrow f$ in measure then a subsequence f_{n_k} converges to f point-wise. By Fataou's lemma

$$\liminf_{n \rightarrow \infty} \int_E f_n \geq \liminf_{k \rightarrow \infty} \int_E f_{n_k} \geq \int_E f$$

If $|f_n| \leq g$, then use Lebesgue dominated convergence theorem to get the same result. □

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a non-negative integrable function. Show that for all $\varepsilon > 0$ there exists a $\delta > 0$ such that for every $E \in \mathcal{E}$ with $m(E) < \delta$,

$$\int_E f < \varepsilon$$

Proof. □

3. (a) Let $A \subset \ell^\infty$ be the set of sequences $a \in \ell^\infty$ such that $\lim_{n \rightarrow \infty} a_n = 0$. Show that A is closed but not compact.

(b) Let $p(x) = (x^2 - 1)^2 - 1$ and fix $b \in A$. Consider the map $F : \ell^\infty \rightarrow \ell^\infty$ given by $F(a)_i := b_i + p(a_i - b_i)$. Show that there exists an $r > 0$ so that if $\|b\|_\infty \leq r$ then the equation $F(x) = x$ has a solution with $x \in A$.

Proof. For part a, consider the sequence $e^{(k)} = (0, 0, \dots, 0, 1, 0, \dots)$ where there is 1 in the k -th place. It is not cauchy so there does not exist a convergent subsequence but the sequence itself is contained in A . □

4. For every $p \in [1, \infty)$ find a function $f \in L^p(\mathbb{R})$ such that $f \notin L^q(\mathbb{R})$ for all $q \neq p$.

Proof. The function

$$\left[\frac{1}{x(\ln x)^2} (\chi_{(0, \frac{1}{2})} + \chi_{(2, \infty)}) \right]^{1/p}$$

works. □

5. Let $f : [-\pi, \pi] \rightarrow \mathbb{R}$ be a continuous function such that $f(-\pi) = 0$ and $f(\pi) = 1$. Let S be the set

$$S := \{n : |\hat{f}(n)| \geq n^{-1.5}\}.$$

Prove that S is an infinite set.

Proof. If S is finite then

$$\left| \sum \hat{f}(n) e^{inx} \right| < \sum |\hat{f}(n)| < \sum n^{-1.5} + \text{error} < \infty$$

Since f is continuous, it equals to its Fourier series but a fourier series cannot have different values on the end-points of a length 2π interval. $\square$

6. Let $D := \{x \in \mathbb{R}^d : \|x\|_2 < 1\}$. Let $f_n : D \rightarrow \mathbb{R}$ be a sequence of continuous functions such that:

- (i) We have $|f_n(x)| \leq (1 - \|x\|_2)^{-1}$ for all $n \geq 1$ and $x \in D$.
- (ii) For every $x \in D$ and $r < 1 - \|x\|_2$ we have

$$f_n(x) = \frac{1}{|B(x, r)|} \int_{B(x, r)} f_n(y) dy$$

where $|B(x, r)|$ is the volume of $B(x, r)$.

Show that there is a subsequence that converges pointwise for all $x \in D$.

7. Let $f \in L^1(\mathbb{R})$ and $g \in L^\infty(\mathbb{R})$. Show that the function

$$x \mapsto \int_{\mathbb{R}} f(x - y)g(y) dy$$

is continuous.

Proof.

$\square$

8. Let (a_n) be a Cauchy sequence in a metric space (M, d) . Show there exists a subsequence (a_{n_k}) such that $d(a_{n_k}, a_{n_{k+1}}) \leq 2^{-k}$ for all $k \geq 1$.

Proof. By definition of being Cauchy.

$\square$

Exercises From Landon's Notes

1. Prove that $L^\infty(E)$ is a complete metric space.